RESEARCH ARTICLE

Using an Insect Mushroom Body Circuit to Encode Route Memory in Complex Natural Environments

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Abstract

Ants, like many other animals, use visual memory to follow extended routes through complex environments, but it is unknown how their small brains implement this capability. The mushroom body neuropils have been identified as a crucial memory circuit in the insect brain, but their function has mostly been explored for simple olfactory association tasks. We show that a spiking neural model of this circuit originally developed to describe fruitfly (*Drosophila melanogaster)* olfactory association, can also account for the ability of desert ants (*Cataglyphis velox)* to rapidly learn visual routes through complex natural environments. We further demonstrate that abstracting the key computational principles of this circuit, which include one-shot learning of sparse codes, enables the theoretical storage capacity of the ant mushroom body to be estimated at hundreds of independent images.

Author Summary

We propose a model based directly on insect neuroanatomy that is able to account for the route following capabilities of ants. We show this mushroom body circuit has the potential to store a large number of images, generated in a realistic simulation of an ant traversing a route, and to distinguish previously stored images from highly similar images generated when looking in the wrong direction. It can thus control successful recapitulation of routes under ecologically valid test conditions.

Introduction

The nature of the spatial memory that underlies navigational behaviour in insects remains a controversial issue, particularly as the neural mechanisms are largely unknown. Insects can perform path integration (PI), using a sky compass and odometer to accumulate velocity into a vector indicating the distance and direction of their start location, typically the nest or hive [1].

研究论文

利用昆虫蘑菇体回路编码复杂自然环境中的路线记忆

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摘要

蚂蚁和许多其他动物一样，会利用视觉记忆沿着复杂环境中的长距离路线行进，但其微小的大脑如何实现这一能力仍不清楚。蘑菇体神经毡被认为是昆虫脑内重要的记忆回路，不过以往对其功能的研究主要集中于简单的嗅觉联想任务。我们发现，最初为解释黑腹果蝇（Drosophila melanogaster）嗅觉联想而建立的蘑菇体脉冲神经毡络模型，也能解释沙漠蚁（Cataglyphis velox）在复杂自然环境中快速学会视觉路线的能力。我们还表明，提炼这一回路的关键计算原理——包括稀疏编码的一次学习——可以估算出蚂蚁蘑菇体在理论上能存储数百幅相互独立的图像。

作者概要

我们提出一个直接依据昆虫神经解剖结构建立的模型，用以解释蚂蚁的循路能力。我们表明，这一蘑菇体回路有潜力存储大量图像；这些图像来自对蚂蚁沿路线行进的逼真模拟。该回路还能将已存储的图像与朝错误方向观察时产生的高度相似图像区分开来。因此，在符合自然生态情境的测试条件下，它能够引导蚂蚁成功重走路线。

引言

昆虫导航行为所依赖的空间记忆究竟具有什么性质，仍存在争议，尤其是其神经机制在很大程度上仍不清楚。昆虫能够进行路径积分（PI）：利用天空罗盘和里程计对速度进行积分，形成一个向量，指示其出发点（通常是巢穴或蜂巢）的距离和方向[1]。

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They are also known to be able to use landmark and/or panoramic visual memories of previously visited locations to guide their movements independently of PI [2,3]. Under normal conditions, both systems are functioning. This raises the possibility that insects additionally store PI vector information with their visual memories [4]; or link their visual memories in sequences [5] or with relative heading vectors [6], forming a topological map; or could even use the PI information to integrate their visual memories into a metric map that represents the spatial relationship of known locations [7].

However another possibility is that PI information is used to determine *which* visual memories to store, for example, the views experienced when facing the nest [8]. Subsequently, such memories can be used directly for guidance without further reference to vector information. Rotating to match the current visual experience with a stored view will give the required heading direction [9], e.g., towards the nest. Surprisingly, this navigation mechanism can exploit multiple memories without necessarily requiring recovery of the 'correct' memory for the current location. Baddeley et al [10,11] presented an algorithm by which an animal attempting to navigate home simultaneously compares the view experienced while it rotates to all memories ever stored while following a PI vector homewards. The direction in which the view looks 'most familiar', i.e., has the best match across all stored views, is generally the correct heading to take to retrace its previous path. In [11] this principle was implemented using the Infomax learning algorithm to train the weights in a two-layer network, where the input is successive images along simulated routes, and the summed activation of the output layer represents the novelty of each image. This implementation was able to replicate many features of ant route following in an agent simulation [11] and has also been shown (with some assumptions about the ant's previous experience) to produce similar search strategies to ants in visual homing paradigms [12].

Strategies of this nature, where the animal does not need to know where it is to know where to go, have been invoked as a more parsimonious explanation for experimental results presented as evidence for a cognitive maps in insects [13]. But is this explanation of navigation plausible, given realistic environmental, perceptual and neural constraints? As pointed out in a recent review [14], parsimony based on what appears simple in computational terms may not map to simplicity with respect to the underlying neural architecture. Yet so far, "nothing is known about neural implementation of navigational space in insects" [14]. In particular, Baddeley et al [11] do not claim that the Infomax implementation of their familiarity algorithm represents the actual neural processing of the ant.

Visual processing in the ant brain has not been extensively studied, but anatomically resembles that of other insects in terms of the initial sensory layers at least. Ants have typical apposition compound eyes, but the size and resolution varies substantially across species. For the desert ants *Cataglyphis* [15] and *Melaphorus bagoti* [16] each eye subtends a large visual field (estimated at 150–170 degrees horizontal extent, with a small frontal overlap) with low visual resolution (interommatidia angles between 3 and 5 degrees). Visual signals pass through the three layers of the optic lobe (lamina, medulla and lobula) maintaining a retinotopic projection but with increasing integration across the visual field [17]. Visual signals then pass directly or indirectly to several other brain regions, including the protocerebrum, the central complex, and the calyxes of the mushroom bodies [17]. The mushroom body (MB) neuropils are conspicuous central brain structures, made up from a large number of Kenyon cells (KCs), whose dendrites together form the calyx and whose axons run in parallel through the penduculus and then bifurcate to form the vertical (or α) and medial (or β) lobes [18]. The extent of visual input to the MB is significantly greater in ants and other hymenoptera than for many other insects for which the MB input is predominantly olfactory [19]. In fact there is increasing evidence that the MB may play a role in visual learning and navigation in insects. There is

研究设计、数据收集与分析、发表决定或稿件准备。

利益冲突：作者声明不存在利益冲突。

已知昆虫还能够利用先前到访地点的地标和 / 或全景视觉记忆，在不依赖路径积分的情况下引导运动[2,3]。通常，两套系统都在运行。因此，昆虫也可能把路径积分向量信息与视觉记忆一同存储[4]；或将视觉记忆按顺序连接[5]，或借相对朝向向量相连[6]，形成拓扑地图；甚至可能利用路径积分信息整合视觉记忆，构成表示已知地点空间关系的度量地图[7]。

另一种可能是，路径积分信息只用于决定应存储哪些视觉记忆，例如朝向巢穴时看到的景象[8]。此后，这些记忆无需再借助向量信息，就可以直接用于导航。只要旋转身体，使当前所见与某个已存储的景象相匹配，就能得到所需的行进方向[9]，例如指向巢穴的方向。出人意料的是，这一导航机制可以同时利用多份记忆，而不一定要找回与当前位置相对应的那一份"正确"记忆。Baddeley 等人[10,11]提出的算法认为，动物在尝试归巢时，会将其旋转过程中看到的景象，同时与此前沿路径积分向量朝家行进时存储的所有记忆比较。看起来"最熟悉"的方向——即与全部已存景象中的某一个匹配得最好的方向——通常就是重走既往路线应采取的正确朝向。文献[11]利用 Infomax 学习算法训练一个双层网络的权重，实现了这一原理：网络输入为模拟路线上的连续图像，输出层的总激活量表示各图像的新异程度。在智能体模拟中，这种实现方式能够再现蚂蚁循路行为的许多特征[11]；在对蚂蚁先前经历作出一定假设的条件下，它也能在视觉归巢实验范式中产生与蚂蚁相似的搜索策略[12]。

这类策略意味着，动物无需知道自己身在何处，也能知道该往哪里走。研究者曾用它来更简约地解释一些被视为昆虫具有认知地图证据的实验结果[13]。但考虑到真实的环境、感知和神经系统约束，这种导航解释是否可信？近期一篇综述指出，从计算角度看似简洁的方案，在底层神经结构上未必同样简单[14]。而截至当时，"对于昆虫导航空间的神经实现方式，我们仍一无所知"[14]。特别是，Baddeley 等人[11]并未声称其熟悉度算法的 Infomax 实现反映了蚂蚁实际的神经处理过程。

蚂蚁脑内的视觉处理尚未得到充分研究，但至少在初级感觉层面，其解剖结构与其他昆虫相似。蚂蚁具有典型的并列式复眼，不过眼睛大小和分辨率在物种之间差异很大。沙漠蚁 Cataglyphis[15] 和 Melaphorus bagoti[16] 的每只眼睛均覆盖宽广的视野（水平方向估计为 150–170°，正前方有少量重叠），但视觉分辨率较低（小眼间夹角为 3–5°）。视觉信号依次经过视叶的三个层次——板层、髓层和视小叶——在保持视网膜位置对应关系的同时，逐层整合更大范围的视野信息[17]。随后，视觉信号直接或间接传向原脑、中央复合体和蘑菇体萼等多个脑区[17]。蘑菇体（mushroom body，MB）神经毡是显著的脑中央结构，由大量肯扬细胞（Kenyon cells，KCs）组成；这些细胞的树突共同形成萼部，轴突则平行穿过柄部，随后分叉形成垂直叶（或 α 叶）和内侧叶（或 β 叶）[18]。与许多主要向蘑菇体输入嗅觉信息的昆虫相比，蚂蚁和其他膜翅目昆虫向蘑菇体输入的视觉信息明显更多[19]。事实上，越来越多的证据表明，蘑菇体可能参与昆虫的视觉学习和导航。已有证据表明，

evidence of expansion or reorganisation of MB at the onset of foraging in ants [20,21] and bees [22–24]. Upregulation of a learning related gene in the MB of honeybees has been linked to orientation flights in novel environments [25]. Cockroaches show impairment on a visual homing task after MB silencing [26], although the central complex rather than the MB appears essential for this task in Drosophila [27]. The MB may nevertheless play a role in some visual learning paradigms in Drosophila [28][29].

To date, the MB have been much more extensively studied in the context of olfactory associative learning, for which they appear crucial [18]. We have previously implemented a spiking neuron model of adult Drosophila MB olfactory learning [30] which used three stages of processing. Olfactory inputs produced a spatio-temporal pattern (an 'image' of the odour) in the antennal lobe (consistent with evidence in flies [31] but also observed in many other insects, including ants [32]). Divergent connectivity from the antennal lobe to the much larger number of KCs that make up the MB project this pattern onto a sparse encoding in a higher dimensional space [33–35]. Reward-dependent learning occurs in the synapses between the KC and a small number of output extrinsic neurons (ENs) [36–38], depending on the delivery at the synapse of an aminergic reward signal [39,40], such that each pattern becomes associated with a positive or negative outcome. This is a simplification of the insect MB circuit, which in reality includes significant feedback connectivity, synaptic adaptation at other levels including in the calyx [41,42] and has a substantial compartmentalisation of its inputs and outputs both between and within the lobes [40]. Nevertheless the basic feedforward architecture implemented in our model was shown to be sufficient to support learning of the association of non-elemental (configural) olfactory stimulus patterns to a reinforcement signal.

Our proposal here is that the MB of the ant allows it to similarly associate visual stimulus patterns, as viewed along a route, with the 'reinforcement' of facing, moving towards or reaching home. In fact the learnt pattern could be multimodal (e.g., including olfactory cues, see Discussion) but we focus here on demonstrating that the ecologically realistic visual learning task posed by route following could be achieved by this circuit architecture. In other words, we suggest the neural architecture of the ant MB could plausibly form the substrate for the familiarity algorithm of Baddeley et al [11]. The complexity of navigation tasks make it difficult to directly measure or manipulate neural circuits in ants under naturalistic conditions to evaluate their contribution to navigation behaviour. Instead, we take a modelling approach, and explore whether our previously developed spiking neural simulation of olfactory learning in the MB of *Drosophila* [30] could be directly applied to the complex ecological task of route memory in ants, using realistic stimuli derived directly from our field experiments.

Results

Reconstruction of the ant's task

We created a realistic reconstruction of the visual experience of ants based on ecologically relevant data from our study of route following in *Cataglyphis velox* [43]. The field site was a flat semi-arid area covered in low scrub and grass tussocks. Ants were trained to forage from a feeder 7.5m from their nest. 15 individual ants were tracked over multiple trips, each revealing an idiosyncratic route to and from the feeder, which they consistently reproduced. We mapped the tussock location and size, and used panoramic pictures taken from ground level to estimate tussock height, and to generate a corresponding virtual environment, where each tussock is a collection of triangular grass blades with a distribution of shading taken randomly from the intensity range in the panoramic pictures (Fig 1). The ground is flat and featureless, and the sky is uniform, without intensity or polarized light gradients. A simulated ant can be placed at any position, with any heading, within this environment. The simulated ant's visual input is

在蚂蚁[20,21]和蜜蜂[22–24]开始外出觅食时，蘑菇体会扩张或重组。蜜蜂蘑菇体内一种学习相关基因的表达上调，与其在新环境中的定向飞行有关[25]。沉默蟑螂的蘑菇体会损害其视觉归巢表现[26]；不过在果蝇中，这项任务所必需的似乎是中央复合体，而非蘑菇体[27]。尽管如此，蘑菇体仍可能参与果蝇的某些视觉学习任务[28,29]。

迄今为止，蘑菇体主要是在嗅觉联想学习的背景下得到深入研究的；它似乎对这类学习至关重要[18]。我们此前建立过成年果蝇蘑菇体嗅觉学习的脉冲神经元模型[30]，其中包含三个处理阶段。嗅觉输入首先在触角叶中形成时空活动模式，即气味的“图像”；这一现象与果蝇中的证据一致[31]，在包括蚂蚁在内的许多其他昆虫中也得到观察[32]。从触角叶到构成蘑菇体的大量肯扬细胞的发散连接，将这一模式投射为高维空间中的稀疏编码[33–35]。依赖奖赏的学习发生在肯扬细胞与少数输出型外在神经元（extrinsic neurons，ENs）之间的突触上[36–38]，其发生取决于突触处是否接收到胺类奖赏信号[39,40]，从而使每种模式与积极或消极的结果建立联系。这是对昆虫蘑菇体回路的简化：真实回路具有显著的反馈连接，在萼部等其他层次也存在突触适应[41,42]，而且各叶之间及叶内部的输入和输出都具有明显的区室化特征[40]。尽管如此，我们的模型表明，其基本前馈结构已足以支持将非元素性（构型性）嗅觉刺激模式与强化信号联系起来的学习。

我们在此提出，蚂蚁的蘑菇体也能以类似方式，将沿路线看到的视觉刺激模式与朝向巢穴、向巢穴移动或抵达巢穴所带来的“强化”联系起来。学习到的模式实际上可能包含多种感觉信息（例如嗅觉线索，见讨论部分）；但本文着重证明，循路行为所要求的、符合真实生态情境的视觉学习任务可以由这一回路结构完成。换言之，我们认为，蚂蚁蘑菇体的神经结构可能为 Baddeley 等人[11]的熟悉度算法提供合理的神经基础。导航任务的复杂性使我们很难在自然条件下直接测量或操纵蚂蚁的神经回路，以评估它们对导航行为的贡献。因此，我们采用建模方法，使用直接源自野外实验的真实刺激，考察此前为果蝇蘑菇体嗅觉学习建立的脉冲神经毡络模拟[30]能否直接用于蚂蚁路线记忆这一复杂的生态任务。

结果

重建蚂蚁的任务

我们依据此前对 Cataglyphis velox 循路行为的研究[43]，利用具有生态真实性的数据重建蚂蚁的视觉体验。野外地点是一片平坦的半干旱区域，覆盖低矮灌丛和草丛。研究者训练蚂蚁从距巢穴 7.5 m 的食物源觅食，在多次往返中追踪 15 只个体；每只蚂蚁都有独特的往返路线，而且会稳定重复。我们测绘草丛的位置和大小，用地面全景照片估计高度，并生成相应虚拟环境：每丛草由三角形草叶组成，其明暗从照片的亮度范围内随机选取（图 1）。地面平坦、无特征，天空均匀，没有亮度或偏振光梯度。模拟蚂蚁可位于环境中任意位置，朝向任意方向。它的视觉输入

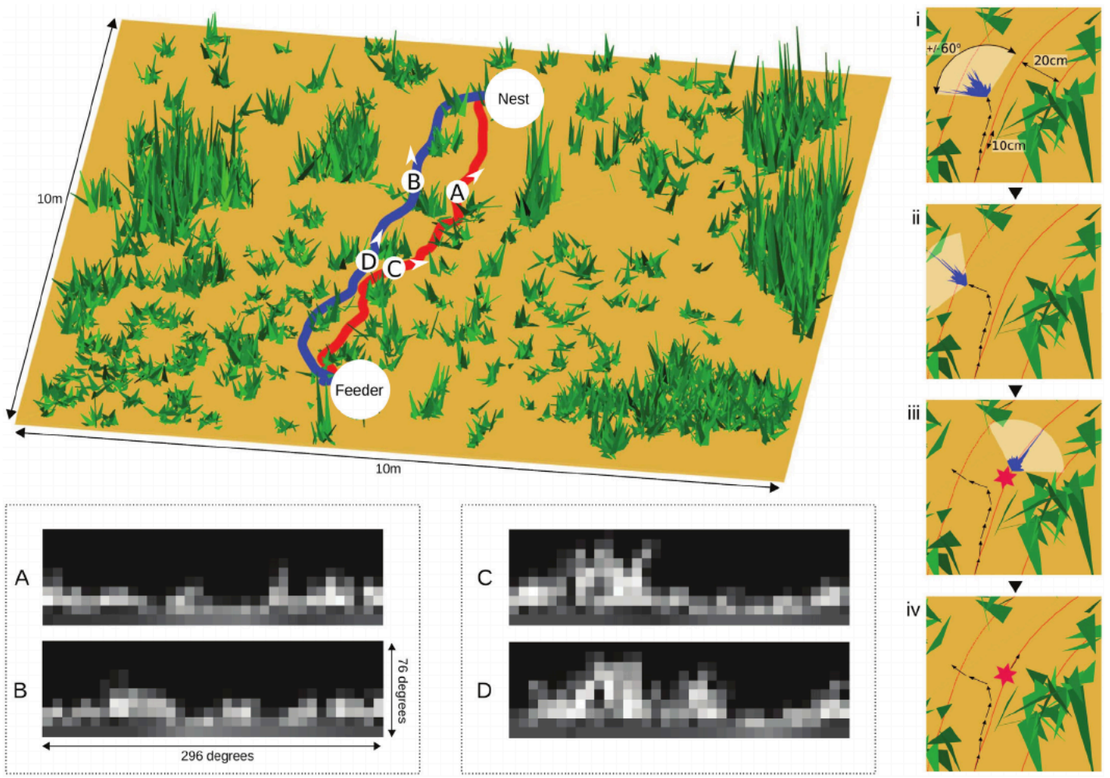


Fig 1. The ant's navigational task. Left: 3D mapping of the real ant environment, which consists of flat ground and clumps of vegetation. Two actual routes followed repeatedly by individual ants from feeder to nest are shown. From a given ground point (e.g. locations A, B, C and D as indicated), the visual input of a simulated ant facing a given direction can be reconstructed, applying a 296 degree horizontal field of view, 8 degree/pixel resolution, inversion of intensity values and histogram equalization. The task of following a specific route requires distinguishing 'familiar views', e.g. for the red route, views A and C, from 'unfamilar views' e.g. B and D from the blue route, despite their substantial similarity. Right: how route following capability is assessed. The simulated ant is trained with images taken at 10cm intervals facing along a route. To retrace the route, it scans +/-60 degrees (i), evaluating the familiarity at each angle (distribution shown in blue), then moves 10 cm in the most familiar direction (ii). Deviating more than 20cm from the route is counted as an error (iii) and the ant is replaced on the nearest point of the route to continue (iv), until home is reached.

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reconstructed from a point 1cm above the ground plane, with a field of view that extends horizontally for 296 degrees and vertically for 76 degrees, with 4 degree/pixel resolution [16], producing a 19x74 pixel image. The image is inverted in intensity and histogram equalized [44] and further down-sampled to 10x36 pixels (effectively 8 degree/pixel resolution) for input to the MB network.

Fig 1 shows the potential difficulty of the route following task within this environment. The ant is surrounded by relatively dense vegetation, which blocks any distant landmarks (see also supplementary S1 Fig which shows a sequence of panoramic photographs taken from ground level along an ant's route, illustrating the lack of any distant features visible throughout the route). The vegetation density is around 2 tussocks/m^2 (compared to 0.05–0.75 tussocks/m^2 used in previous simulations [11]) and ants are often observed to go directly through tussocks leading to abrupt changes in the view. As a consequence, multiple non-overlapping views need

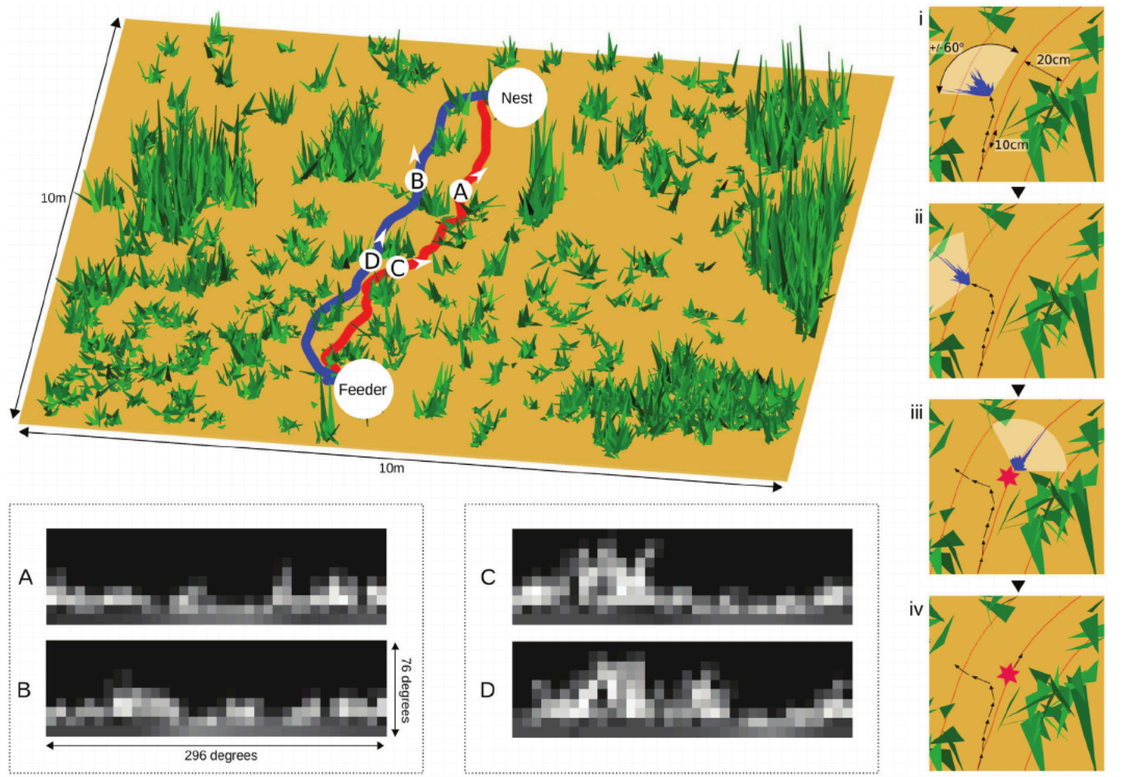


图 1 | 蚂蚁的导航任务。左：真实蚂蚁活动环境的三维地图，由平坦地面与成丛植被构成；图中显示不同蚂蚁从食物源到巢穴时反复行走的两条真实路线。在给定地面位置（如 A、B、C、D），可重建模拟蚂蚁朝某一方向观察时的视觉输入：水平视野 296°、每像素 8°，并进行亮度反转与直方图均衡化。循指定路线时，必须区分“熟悉视景”（如红色路线的 A、C）与高度相似的“不熟悉视景”（如蓝色路线的 B、D）。右：循路能力的评估。训练时，沿路线每隔 10 cm 采集一幅朝向路线前进方向的图像。重走路线时，蚂蚁扫描 ±60°（i），评估各角度的熟悉度（蓝色分布），再朝最熟悉方向前进 10 cm（ii）。偏离路线超过 20 cm 计为一次错误（iii），随后将其移回路线最近点继续（iv），直至到家。

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其视觉输入从离地面 1 cm 的位置重建：水平视野为 296°，垂直视野为 76°，分辨率为每像素 4°[16]，得到 19×74 像素的图像。图像经亮度反转和直方图均衡化[44]后，进一步降采样为 10×36 像素（实际相当于每像素 8°），再输入蘑菇体（MB）网络。

图 1 展示了蚂蚁在这一环境中循路可能面临的困难。蚂蚁周围的植被较茂密，遮挡了远处所有地标（另见补充图 S1：沿蚂蚁路线从地面拍摄的一系列全景照片，表明整条路线都看不到远处特征）。植被密度约为每平方米 2 丛草，而此前模拟采用的是每平方米 0.05–0.75 丛[11]。研究者还经常观察到蚂蚁直接穿过草丛，导致视景骤变。因此，需要多个互不重叠的视景

to be stored to encode the full route, and searched when recapitulating it. The views lack unique features, especially at the ant eye's low resolution, so the potential for aliasing seems high. In particular, there is no reason to expect images seen by the ant along one route (e.g. Fig 1A and 1C) to have any common properties that could be learnt to distinguish them from non-route images (e.g. those from another ant's route, Fig 1B and 1D; or those experienced when not aligned with the route).

Mushroom body processing

We altered our mushroom body model [30] (Fig 2) only by increasing the number of neurons (staying well below estimates for the ant brain [17]), and introducing anti-Hebbian learning [39,40]. Our input layer consists of 360 visual projection neurons (vPNs), activated proportionally to the intensity of the pixels in a scaled and normalized image (Fig 1). We intentionally kept this visual pre-processing simple, as there is little evidence on which to make assumptions about the nature of the visual input to the MB in the ant. The second layer contains 20,000—KCs, each receiving input from 10 randomly selected vPNs [35,45] (note this does not preserve retinotopy), and needing coincident activation from multiple vPNs to fire. This allows decorrelation of images using a sparse code [46], i.e., only a few KCs (around 200) will be activated, and the activation patterns will be more different than the input patterns. All KC outputs converge on a single extrinsic neuron (EN) with a three-factor rule for learning [47]. This uses the relationship of presynaptic and postsynaptic spike timing to 'tag' synapses, and a global reinforcement signal to permanently decrease the strength of tagged synapses, consistent with neurophysiological evidence from the MB of locusts [39]. Thus, images that have been previously paired with reinforcement will excite KCs that no longer activate EN as these connections have been weakened.

We assume that the ant learns a homeward route by storing the views encountered as it follows its path integration home vector back to the nest. The 'reinforcement' signal could thus be generated by decreases in home vector length. In practice, we train the network by generating the images that would be seen every 10cm along the ~8m recorded route of a real ant, with the heading direction towards the next 10cm waypoint. These 80 images are each presented (statically) to the network for 40ms, followed by a transient reinforcement signal (Fig 3). The parameters in the model are set to effectively result in 'one shot' learning given this timing of presentation: i.e., a single pairing of image and reinforcement causes the relevant synaptic weights to be reduced to near zero. Such learning is not implausible as it has been shown that, for example, individual bees can acquire odour associations in one or two trials [48]. After training, the ant should be able to recover the heading direction at any point along the route by scanning and choosing the 'most familiar' direction as indicated by the minima in the activity of EN during the scan.

We compare the MB model to two alternatives. 'Perfect memory' represents the best possible performance, by assuming the ant photographically stores all 80 images and directly compares the current viewpoint with all stored images to find the highest similarity, i.e., the minimum in the sum of squared pixel intensity differences (see Methods). The Infomax algorithm, used previously for this task [11] but under less realistic environmental and perceptual processing constraints, attempts to build a generative model of the 80 views using a fully connected two layer neural network (see Methods). We note that a much higher learning rate was needed in the current study to get successful results from Infomax, and it is possibly learning by overfitting, i.e., its 'model' consists essentially of the 80 presented views. In Fig 4A–4C we show the directional choice that would be made using the output from each method for a short segment of the path, using displacements at 5cm intervals up to +/-25cm away from the

被存储，以编码完整路线，并在重走时进行检索。这些视景缺少独有特征，尤其在蚂蚁眼睛的低分辨率下，发生视景混淆的风险似乎很高。尤其没有理由预期，蚂蚁沿同一路线看到的图像（如图 1A、1C）具有某些共同属性，能够通过学习与路线外的图像区分开；后者包括另一只蚂蚁路线上的图像（图 1B、1D），或朝向不与路线一致时看到的图像。

蘑菇体的信息处理

我们对先前的蘑菇体模型[30]只作了两项修改：增加神经元数量（仍远低于对蚂蚁脑的估计值[17]），并引入反赫布学习[39,40]（图 2）。输入层由 360 个视觉投射神经元（vPN）组成，其激活程度与经缩放、归一化处理的图像像素亮度成正比（图 1）。由于缺少充分证据来推断蚂蚁蘑菇体所接受的视觉输入形式，我们有意采用了简单的视觉预处理。第二层含 20,000 个肯扬细胞（KC），每个细胞从随机选取的 10 个 vPN 接收输入[35,45]；这些连接不保持视网膜拓扑关系，而且 KC 只有在多个 vPN 同时激活时才会放电。这使图像通过稀疏编码实现去相关[46]：每幅图像仅激活少数 KC（约 200 个），不同图像的激活模式比输入模式更易区分。所有 KC 的输出汇聚至单个外在神经元（EN），并按三因素规则学习[47]。该规则利用突触前、后放电时间的关系为突触加“标记”；当全局强化信号到来时，被标记突触的强度便永久降低。这与蝗虫蘑菇体的神经生理证据一致[39]。因此，先前与强化信号配对的图像虽会激活 KC，但由于相应连接已减弱，这些 KC 不再激活 EN。

我们假定，蚂蚁沿路径积分所得的返巢向量回巢时，会存储沿途所见视景，从而学会返巢路线。返巢向量长度的减小可能产生这里的“强化”信号。实际训练中，我们沿一只真实蚂蚁约 8 m 长的记录路线，每隔 10 cm 生成一幅朝向下一个路点的图像。80 幅图像各静态呈现 40 ms，随后给予一个短暂的强化信号（图 3）。模型参数设为可在这一呈现时序下实现“一次学习”：图像与强化信号只需配对一次，就可使相关突触权重降至接近零。这样的学习并非不可想象，例如单只蜜蜂可在一两次试验中学会气味联想[48]。训练后，蚂蚁应能在路线任一点通过扫描寻找 EN 活动的最低值，以此确定“最熟悉”的方向并恢复行进朝向。

我们将蘑菇体模型与两种替代方法比较。“完美记忆”代表可能达到的最佳表现：假设蚂蚁如同拍照一样存储全部 80 幅图像，将当前视景与所有记忆直接比较，找出最高相似度，即像素亮度差平方和的最小值（见“方法”）。曾用于这一任务的 Infomax 算法[11]以全连接双层网络为 80 个视景建立生成模型（见“方法”），但此前的环境和知觉处理约束较不逼真。本研究中，Infomax 必须使用高得多的学习率才会成功；它可能通过过拟合学习，也就是“模型”实质上由所呈现的 80 个视景本身构成。图 4A–C 显示各方法在一小段路径上的方向选择；测试点相对存储位置按 5 cm 间隔取样，位移最大为 ±25 cm。

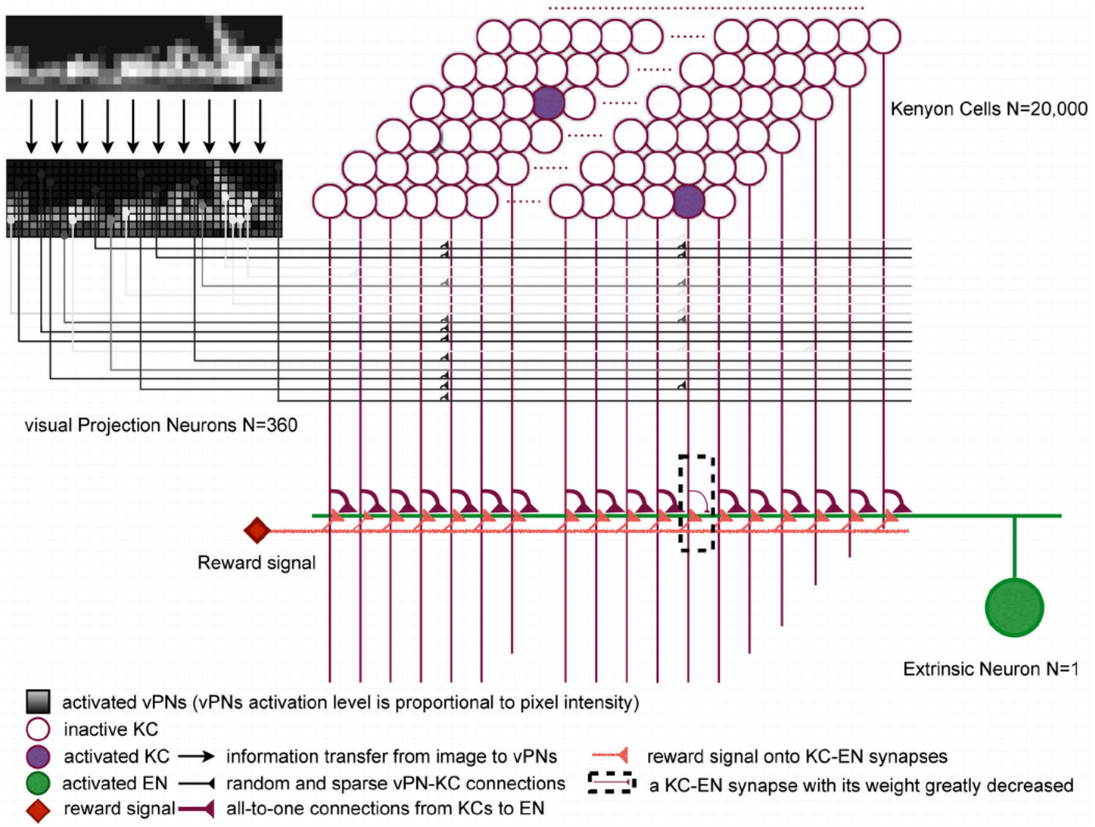


Fig 2. The architecture of the mushroom body (MB) model. Images (see Fig 1) activate the visual projection neurons (vPNs). Each Kenyon cell (KC) receives input from 10 (random) vPNs and exceeds firing threshold only for coincident activation from several vPNs, thus images are encoded as a sparse pattern of KC activation. All KCs converge on a single extrinsic neuron (EN) and if activation coincides with a reward signal, the connection strength is decreased. After training the EN output to previously rewarded (familiar) images is few or no spikes.

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locations at which images were stored. The MB model produces directional output of equivalent reliability to the other visual memory methods, pointing the simulated ant along the route with only a few exceptions. Note that a perfect match is possible only if the simulated ant is in exactly the same location as the memory was stored, but all the methods are quite robust for nearby locations.

Evaluating route following

We simulate retracing the route starting from the feeder, heading along the route (Fig 1, right). The next heading direction is determined by the minima in EN firing (or the equivalent choice made using Perfect Memory or Infomax) for a directional scan of ±60 degrees (reflecting a general bias to continue in the same direction). A 10cm step is taken in this direction and the process repeated until the ant reaches home. Note that it is thus possible for the simulated ant to

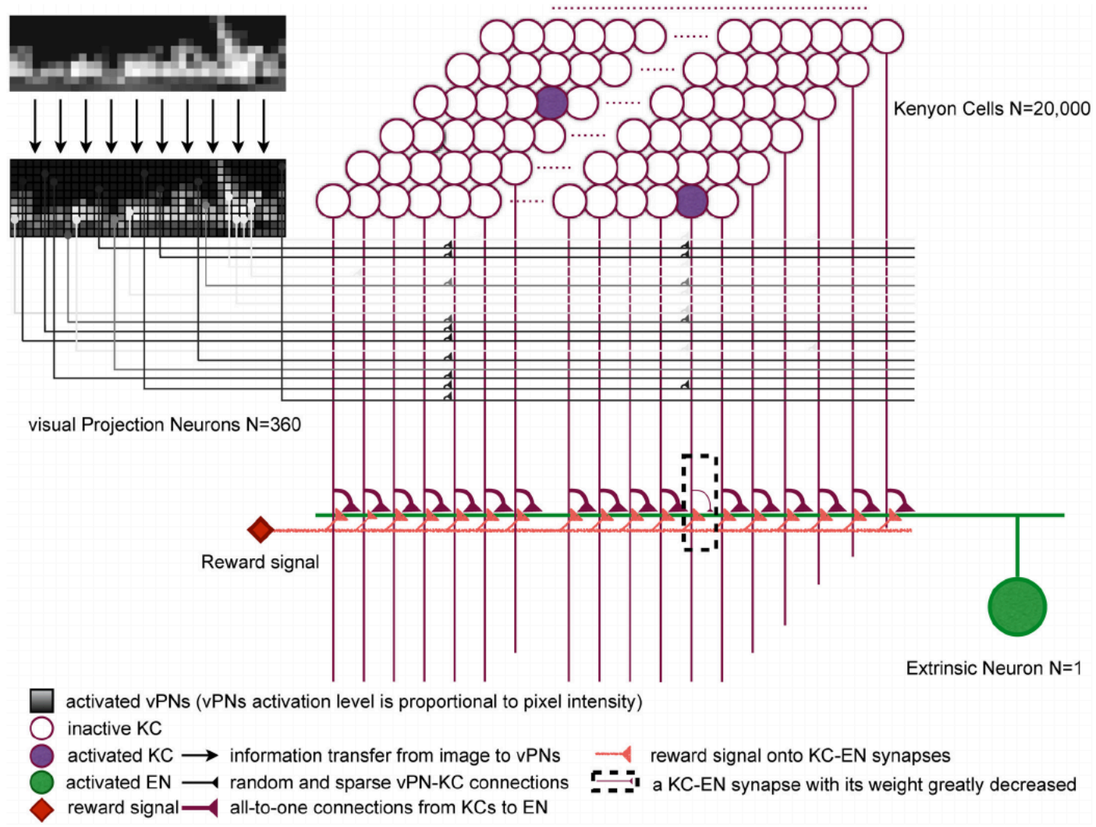


图 2 | 蘑菇体（MB）模型的结构。图像（见图 1）激活视觉投射神经元（vPN）。每个肯扬细胞（KC）接收随机选取的 10 个 vPN 输入，只有多个 vPN 同时激活时才超过放电阈值；图像由此被编码为稀疏的 KC 活动模式。全部 KC 汇聚到单个外在神经元（EN）；若激活与奖赏信号同时发生，连接强度就下降。训练后，EN 对先前获得奖赏的熟悉图像只产生很少脉冲或不放电。

doi:10.1371/journal.pcbi.1004683.g002

这些测试点以存储图像的位置为基准。蘑菇体模型的方向输出与其他视觉记忆方法同样可靠，除少数例外，都指引模拟蚂蚁沿路线前进。注意，只有当模拟蚂蚁与记忆存储位置完全重合时，才可能得到完美匹配；但所有方法对附近位置都相当稳健。

循路表现的评估

我们模拟蚂蚁从食物源出发沿路线重走返巢（图 1 右）。在 ±60° 内进行方向扫描，按 EN 放电的最低值（或完美记忆、Infomax 的相应选择）决定下一步朝向，这体现了总体上继续向原方向行进的倾向。蚂蚁沿该方向走 10 cm，再重复过程直至到家。因此，模拟蚂蚁可能

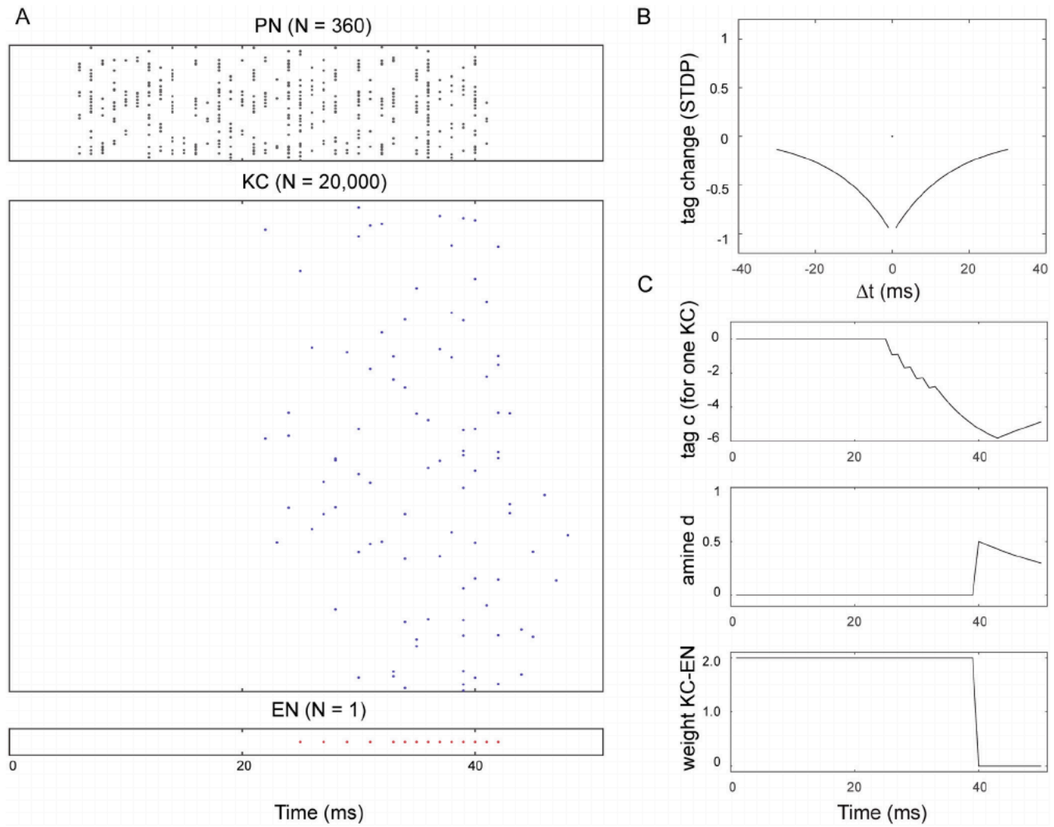


Fig 3. The response of the network during training with one image. A: The image is presented for 40ms, directly activating the vPNs which respond with a spiking rate proportional to the intensity of their input pixel. This produces sparse activation of the KC, which causes the EN to fire. B: An STDP process tags KC synapses depending on the relative timing, Δt, of their spikes to spikes in EN. C: Within 40ms, an active KC will have a strongly negative tag. An increase of amine d, representing reinforcement, will combine with the tag to greatly reduce the weight of the KC-EN synapse.

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be scanning from a slightly different position from where memory was stored, and hence for the wrong direction to be chosen, even for perfect memory. If successive movements lead the ant a significant distance from the route (> 20cm) then, in the cluttered environment we are testing, matches become poor and the ant will pursue a random course with little chance of recovery. Hence, if a step results in a location more than 20cm from the route, the error count is increased by one and the ant is placed back on the nearest point on route. We count the total number of such errors that occur before the ant comes within 20cm of the home position. As a baseline, we include a random control in which the visual information is ignored and the direction on each step is chosen randomly from ±60 degrees.

The performance of the model is assessed using 15 different ~8m routes, based on the observed routes of real ants recorded in our field study. The results are shown in Fig 4. Random directional choice produces a mean of 18.7 errors (standard deviation = 3.6) per route. Perfect memory has a much lower number of errors (mean 1.1, s.d. = 0.9) suggesting familiarity is an

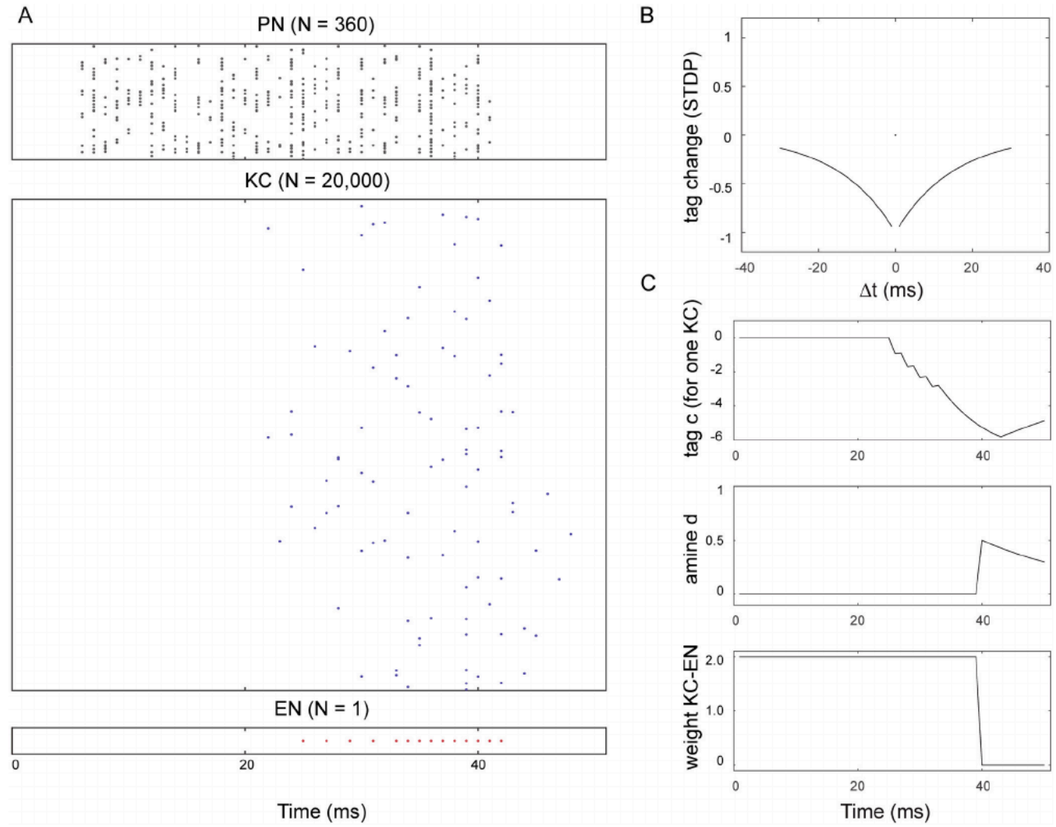


图 3 | 用一幅图像训练时的网络反应。A：图像呈现 40 ms，直接激活 vPN，其放电频率与输入像素亮度成正比；这使 KC 稀疏激活，并引发 EN 放电。B：STDP 过程按 KC 相对 EN 的脉冲时差 Δt 标记 KC 突触。C：40 ms 内，活跃 KC 的标记变为显著负值；代表强化的胺浓度 d 上升，与标记共同作用，大幅降低 KC–EN 突触权重。

doi:10.1371/journal.pcbi.1004683.g003

在略偏离记忆存储位置的地点扫描，即使使用完美记忆，也可能选错方向。在本文测试的杂乱环境中，如果连续移动使蚂蚁明显偏离路线（超过 20 cm），视景匹配就会变差，蚂蚁会随机行进，几乎没有恢复的机会。因此，每当一步移动使它距路线超过 20 cm，就将错误数加一，并把它移回路线上的最近点。我们统计蚂蚁进入距巢穴位置 20 cm 范围前的错误总数。作为基线，随机对照忽略视觉信息，在每一步的 ±60° 内随机选向。

我们依据野外真实蚂蚁的记录路线，用 15 条不同的约 8 m 路线评估模型，结果见图 4。随机选向每条路线平均产生 18.7 次错误（标准差 3.6）。完美记忆的错误数低得多（均值 1.1，标准差 0.9），说明熟悉度是一种

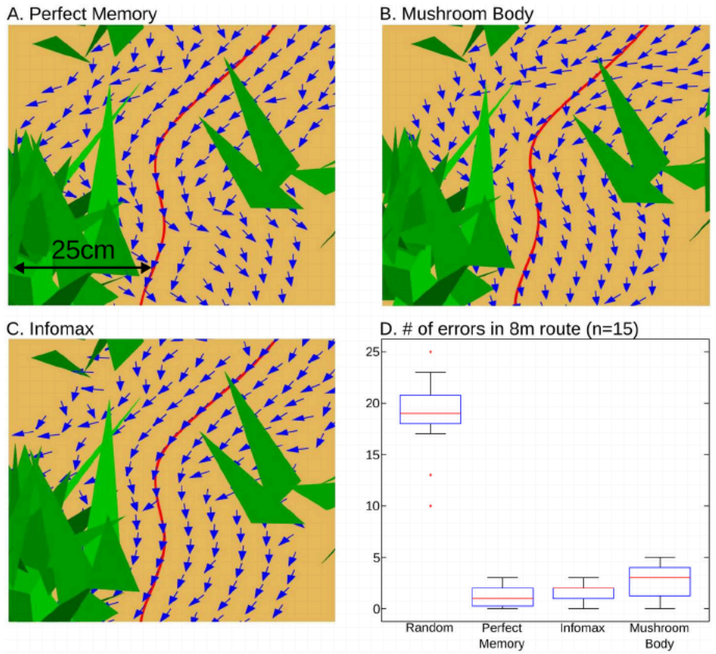


Fig 4. Performance of the familiarity algorithm. A-C: Perfect memory (A), the mushroom body (B) and infomax (C) are evaluated for a segment of the route (triangles are grass blades) after training with ~80 images along this route. For test locations in 5cm displacements up to 25cm away from the trained image locations (on red line), all three familiarity algorithms are robust, recovering directions (arrows) that enable route following, but even 'perfect' memory can produce errors when not tested at an identical location to where the image was stored. D: Comparison of number of errors made by each algorithm when retracing a route (see Fig 1), compared to random choice of direction. Boxplots show the median, interquartile range and maximum and minimum results for 15 different routes, each ~8m long, with images stored every 10cm.

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effective route following method. The MB model is only slightly less successful than this benchmark, with a mean of 2.6 errors (s.d. = 1.5). This is significantly worse than Perfect Memory (Bayes factor, calculated using the method in [49], is 28:1 in favour of non-equal means), but the drop in performance is small (1.5 additional errors) given the substantial reduction in computational demand – Perfect Memory requires separate comparison of the test view with every stored view for every direction in a scan, whereas MB simply uses the immediate output of the network. Infomax produces a mean of 1.5 errors (s.d. = 0.8), which is marginally better than MB (Bayes factor 2.8:1 in favour of non-equal means). But note that compared to MB, Infomax uses at least 5 times more synaptic weights, and needs to adjust every weight in the network for every training input, using a non-local rule, which is biologically less feasible. S1 Movie shows a route and images corresponding to a run in which MB produced no errors.

Analytical solution for storage capacity of the mushroom body circuit

The key properties of the MB circuit for this task are the small number of neurons activated in the KC layer (around 1% of neurons activated by each image) and fast (one-shot) learning with

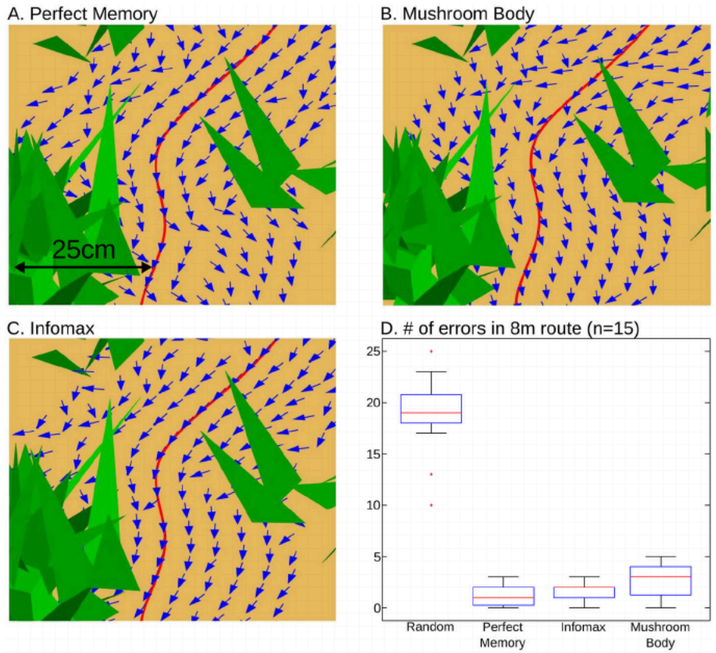


图 4 | 熟悉度算法的表现。A–C：沿路线用约 80 幅图像训练后，在一小段路线（草叶为三角形）上评估完美记忆（A）、蘑菇体（B）与 Infomax（C）。测试点相对训练图像位置（红线）以 5 cm 为间隔，最远偏离 25 cm。三种方法都能稳健恢复支持循路的方向（箭头），但测试与存储位置不完全相同时，即使完美记忆也可能出错。D：各算法与随机选向重走路线（见图 1）时的错误数。箱线图表示 15 条约 8 m 路线的中位数、四分位范围、最大值和最小值；图像每隔 10 cm 存储一次。

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有效的循路方法。蘑菇体模型略逊于这一基准，平均 2.6 次错误（标准差 1.5）。其表现显著差于完美记忆：按文献[49]的方法计算，支持均值不相等的贝叶斯因子为 28:1。不过，考虑到计算需求的大幅下降，性能只少了每条路线 1.5 次错误。完美记忆在每次扫描的每个方向上，都要将测试视景逐一与所有已存视景比较；蘑菇体则直接使用网络的即时输出。Infomax 平均 1.5 次错误（标准差 0.8），比蘑菇体略好，两者均值不相等的贝叶斯因子为 2.8:1。但与蘑菇体相比，Infomax 至少使用 5 倍的突触权重，而且每个训练输入都需要按非局部规则调整全部权重，生物学可行性较低。补充视频 S1 展示了一次蘑菇体模型无错误的循路过程和相应图像。

蘑菇体回路存储容量的解析解

这一任务中，蘑菇体回路的关键性质是 KC 层激活的神经元很少（每幅图像约激活 1%），并且能够快速（一次）学习，

no forgetting. We can abstract the spiking KCs as nodes with a binary state, representing whether or not they activated above threshold by the input pattern, and also abstract the KC-EN synaptic strength as either high (contributes to response in EN, the initial state) or low (no input to EN, the state after unidirectional, rapid learning). Because of the initially high setting (if KC is active, EN will fire), we can also simplify to a two factor learning rule: if KC activation coincides with reward, set the synapse strength permanently to low. This abstraction essentially views the learning network of the MB as a layer of binary units with outputs converging on one output unit via binary synapses that have a unidirectional plasticity rule.

Such a network is comparable to a Willshaw net [50] for which theoretical estimations of information capacity for a given number of input units N has been previously examined within a framework of fixed sparseness, binomially distributed [51]. However, the activity of each KC is not completely independent because all KCs sample from the same population of vPNs activity patterns at any given time and thus the probability distribution for the number of active KCs is not a binomial distribution [52]. Nevertheless, the expected number of active KCs is Np_{kc} (where N is the number of KCs and p_{kc} the probability of a connection between each vPN and KC) as required, while the correlation between KCs can be reduced by changing the number of vPNs sampled by each KC and thus the capacity estimation with binomial approximation may hold. Similar capacity estimations have been previously performed for networks with binary synapses and bidirectional plasticity, showing that sparseness prolongs memory lifetimes by reducing the rate of plasticity [53] and therefore the interference between new memory encoding and stored memories. Note, in our MB model, increased sparseness also influences plasticity rates and thus changes in capacity can be seen as variations in the interference of the learned patterns.

The abstracted MB allows theoretical estimation of the memory capacity m (the number of patterns that can be stored) of the mushroom body for a given size N (number of KC) and average activity p (average proportion of KC activated by each pattern). We derived (see Methods) an expression for the mean number of patterns m that can be stored before the probability of error reaches P_{error}, where error is defined as having a random unlearned pattern that activates only KC nodes that already have their KC-EN weight set low, thus producing the same EN output as a learned pattern. The resulting capacity is given by:

$$m = \frac{ln\left(\frac{1-P_{error}^{\frac{1}{N}}}{p}\right)}{ln(1-p)} \quad (1)$$

The storage capacity as a function of network size and sparsity is shown in Fig 5. Given the above assumptions, our MB model, with $N = 20{,}000$ input units, one output unit, and average KC activity $p = 0.01$ should allow around 375 random images to be stored before the probability of confusing a novel image with a stored image exceeds $P_{error} = 0.01$. We confirmed this by training our MB network with random KC patterns, and testing with 100 novel patterns. It was indeed the case that more than 350 patterns could be stored before any of the novel patterns were mistakenly classed as familiar (i.e. produced no spikes in EN, see S2 Fig). Following the same procedure for varying network size and average KC activity also produces results comparable with the theoretical predictions (Fig 5, diamonds). If memories are stored every 10cm as we have assumed, memorising 350 patterns corresponds to an ant being able to recall a route of 37.5m, or several routes of around 10m, before any confusion would occur; in uncluttered environments, memories could be more spaced (e.g. every 1m or more) and distances correspondingly increased (routes of hundreds of metres, [54]). The actual upper limit of ant route memory has not been systematically explored but these values are on the same

且不发生遗忘。我们可以把产生脉冲的肯扬细胞（KC）抽象为具有二值状态的节点，表示输入模式是否使其活动超过阈值；也可以把 KC–EN 突触强度抽象为高、低两种状态：高强度是初始状态，会对外在神经元（EN）的反应作出贡献；低强度是经过单向、快速学习后的状态，不再向 EN 提供输入。由于初始权重较高（只要 KC 活跃，EN 就会放电），我们还可以把学习规则简化为两因素规则：若 KC 激活与奖赏同时发生，就将突触强度永久设为低值。这一抽象实际上把蘑菇体学习网络视为一层二值单元，其输出经具有单向可塑性规则的二值突触，汇聚到一个输出单元。

这种网络类似于 Willshaw 网络[50]；先前已有研究在稀疏度固定、活动服从二项分布的框架下，估计给定输入单元数 N 时的信息容量[51]。然而，各 KC 的活动并非完全独立，因为任一时刻所有 KC 都从同一组 vPN 活动模式中取样，所以活跃 KC 数量的概率分布不服从二项分布[52]。尽管如此，活跃 KC 的期望数量仍按要求为 Np_{kc}（其中 N 是 KC 数量，p_{kc} 是每个 vPN 与 KC 之间的连接概率）。（译注·存疑：此处 p_{kc} 的定义可能混淆了“连接概率”与“激活概率”。$N{\cdot}p_{kc}$ 表示活跃细胞数的期望时，p_{kc} 应为单个 KC 的激活概率；仅凭 vPN–KC 连接概率不能直接得出该期望。参见本文第 17 页对活动概率 p 的定义；尚未核实是否存在正式勘误。）通过改变每个 KC 取样的 vPN 数量，可降低 KC 之间的相关性，因此以二项分布近似得到的容量估计可能仍适用。先前对具有二值突触和双向可塑性的网络也做过类似估计：稀疏活动降低可塑性发生率，从而延长记忆寿命[53]，也减弱新记忆编码与已有记忆的干扰。注意，在我们的蘑菇体模型中，提高稀疏度同样影响可塑性发生率，因此容量变化也可视为已学习模式间干扰程度的变化。

抽象后的蘑菇体模型可用于理论估计其记忆容量 m（可存储模式数）；给定条件为网络规模 N（KC 数量）与平均活动比例 p（每种模式平均激活的 KC 比例）。我们推导了一个表达式（见“方法”），给出错误概率达到 P_{error} 前平均能够存储的模式数 m。这里的错误指：一个未经学习的随机模式，只激活那些 KC–EN 权重已经被设为低值的 KC 节点，从而产生与已学习模式相同的 EN 输出。得到的容量如下：

$$m = \frac{ln\left(\frac{1-P_{error}^{\frac{1}{N}}}{p}\right)}{ln(1-p)} \quad (1)$$

图 5 展示了存储容量随网络规模与稀疏度的变化。在上述假设下，我们的蘑菇体模型含 N=20,000 个输入单元、一个输出单元，平均 KC 活动比例 p=0.01，应可存储约 375 幅随机图像，之后把新图像与已存图像混淆的概率才会超过 P_{error}=0.01。我们用随机 KC 模式训练网络，并以 100 个新模式测试，验证了这一点。确实，在任何新模式被误判为熟悉模式（即 EN 不放电，见补充图 S2）之前，可存储 350 多个模式。改变网络规模和平均 KC 活动水平后采用相同程序，也得到与理论预测相近的结果（图 5 中的菱形）。如果如本文假设每隔 10 cm 存储一次记忆，那么记住 350 个模式，就相当于蚂蚁能在不发生混淆之前记住一条 37.5 m 的路线，或数条约 10 m 的路线。（译注·已核实：模式数与距离的换算不一致。按每个模式约对应 0.1 m 估算，350×0.1 m≈35 m；37.5 m 则对应约 375 个模式。作者究竟希望更正模式数还是距离，本文未说明，故保留原文两项数值。）在较开阔的环境中，记忆点可以间隔更远（如每隔 1 m 或更远），相应距离也会增加（可达数百米[54]）。蚂蚁路线记忆的实际容量上限尚未得到系统研究，但这些数值与大多数实验研究所用的数值处于相同

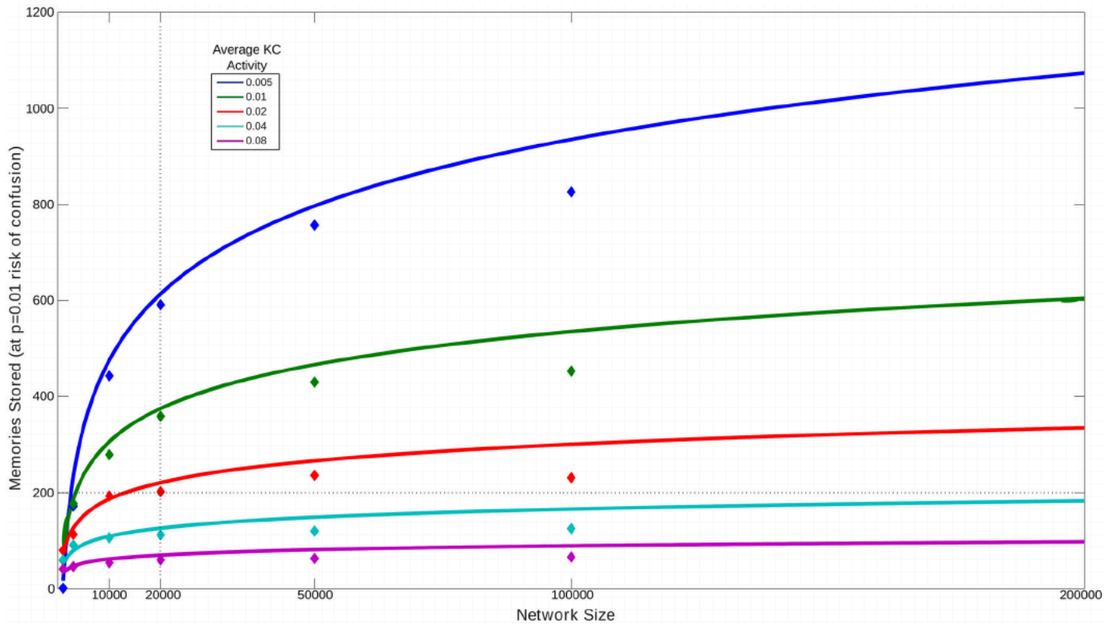


Fig 5. The number of independent images that an abstracted MB network can store before the probability of an error (producing an output of 0 for a novel image) exceeds 0.01. Lines: predictions from theoretical analysis. Diamonds: results from equivalent simulations using the full spiking model. The capacity scales logarithmically with the number of neurons, and increases if fewer KCs are activated on average by each pattern.

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order as those used in most experimental studies (e.g. [43,55,56]). There are also a number of plausible ways in which the capacity could be increased: e.g., having more than one EN; more states for synapses; probabilistic rather than deterministic synapse switching; or preprocessing the image data.

To compare further the capacity estimate derived from this abstraction (with independent random input patterns) to the practical performance of the MB network (with correlated input patterns from routes) we ran the following test. The MB was trained successively with each image from every route, to a total of 1200 images. After each image was added to memory, the EN response was recorded i) for that image, ii) for an image taken 5cm away and facing the same direction, iii) for an image taken from a random location in the ant environment, and iv) for a completely random image. Distinguishing i) from ii-iv) means that stored memories are not confused with new images. Distinguishing ii) from iii) is helpful for robust route following, i.e., the right direction from small displacements should still look more familiar than random locations. Distinguishing i-iii) from iv) might be expected because completely random images will rarely look like images from the environment, where sky is always above grass, which is above the ground, etc. As shown in Fig 6, the response of EN is very noisy (plotted points), hence occasional mistakes in familiarity will occur, but nevertheless the response for stored images is on average (fitted curves) clearly distinguishable from other images as it produces no EN spikes, and images from nearby locations tend to be more familiar than images from random locations, even after storing 1200 images.

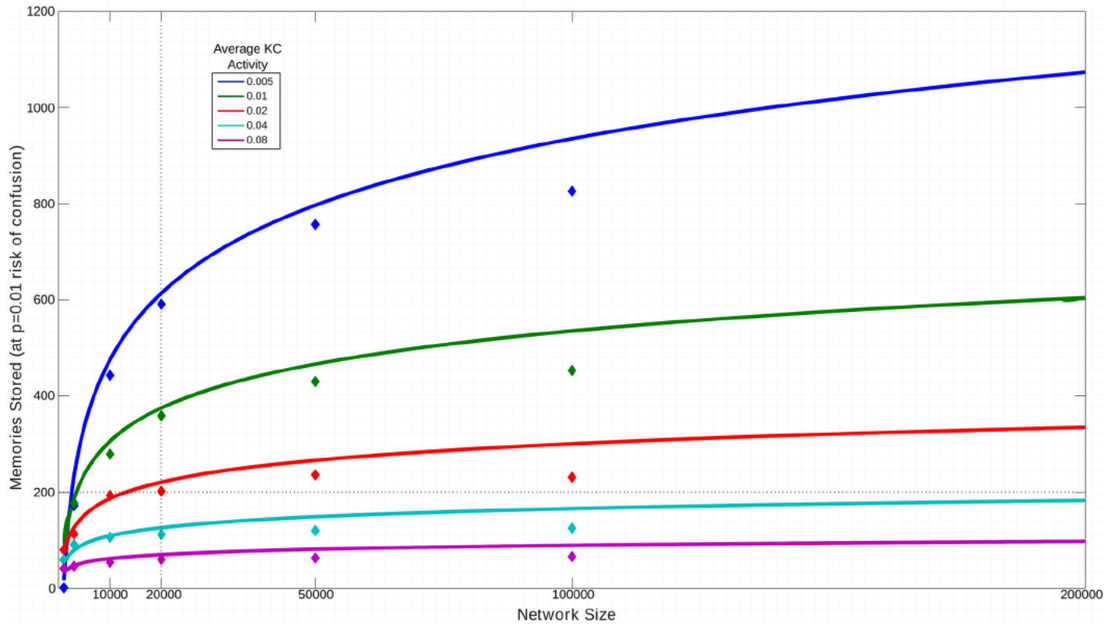


图 5 | 抽象蘑菇体网络可存储的独立图像数量；超过该容量，将新图像误判为熟悉（输出零）的概率就超过 0.01。曲线：理论分析预测；菱形：完整脉冲模型的相应模拟。容量随神经元数量呈对数增长；每种模式平均激活的 KC 越少，容量越大。

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数量级（例如[43,55,56]）。还有若干合理方式可以增加容量，例如使用多个 EN、增加突触状态数、以概率性而非确定性方式切换突触状态，或预处理图像数据。

为了进一步比较由独立随机输入模式推得的容量估计与采用相关路线图像时蘑菇体网络的实际表现，我们依次用每条路线的图像训练网络，总计 1200 幅。每加入一幅图像后，记录 EN 对四类图像的反应：①刚存入的图像；②在距该位置 5 cm 处、朝向相同的图像；③蚂蚁环境中随机位置的图像；④完全随机的图像。若能区分①与②–④，说明已存记忆不会与新图像混淆；若能区分②与③，则有助于稳健循路，即使稍微偏离路线，正确方向仍应比随机位置更熟悉。环境中的天空总在草丛上方、草丛又在地面上方，因此完全随机的图像很少与环境图像相似；据此也可预期①–③与④能够区分。图 6 显示，EN 响应有较大噪声（散点），因此偶尔会发生熟悉度判断错误。尽管如此，平均响应（拟合曲线）仍表明，已存图像使 EN 完全不放电，明显区别于其他图像；即使已存储 1200 幅图像，附近位置的图像通常仍比随机位置的图像更熟悉。

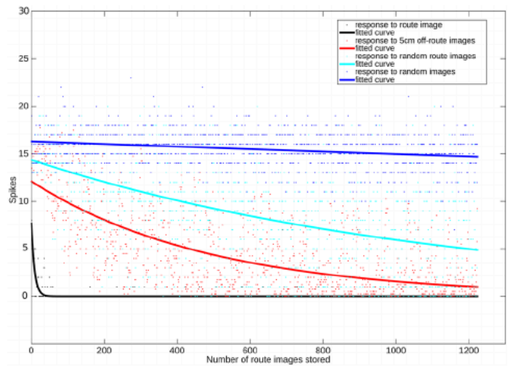


Fig 6. Testing the capacity of the MB network to distinguish familiar from novel images as additional route images are stored (x-axis). On average (fitted curves), the EN output to learned images is zero. Images from nearby locations with the same heading are more familiar (lower EN response) than those from random locations. Random images remain clearly distinguishable even with 1200 images stored.

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Discussion

We have shown that the neural architecture of the insect mushroom body can implement the 'familiarity' algorithm [11] for ant route following. In our simulation, the MB learns over 80 on-route images, reconstructed from real ant viewpoints in their natural habitat, and reliably distinguishes them from the highly similar images obtained when looking off-route. This is the first model of visual navigation in insects to draw such a close connection to known neural circuits, rather than appealing to abstract computational capacities; as a result we found that a simpler associative learning network (computationally equivalent to a Willshaw net [50]) than Infomax suffices for the task.

Strikingly, the network which was originally developed for modelling simple olfactory associative learning in flies required almost no modification other than increased size to perform the apparently much more complex function of supporting navigation under realistic conditions. Ants (and other navigating insects such as bees) are known to have significantly enlarged MB compared to flies [57]. Visual memory has also been localised to the central complex of the insect brain [27,58], and it remains possible that navigation, including visual memory for homing routes, is carried out entirely in the central complex and does not involve the MB. We would agree that the central complex is likely to be a key area for navigational capabilities in many insects [27,59], in particular processing polarised light [60], and potentially for path integration. However the extreme memory demands of extended route following in hymenoptera such as ants and bees may have co-opted the unique circuit properties of the MB: a divergent projection of patterns into a sparse code across a very large array of neurons, enabling separation and storage of a large number of arbitrary, similar patterns. Such an architecture appears to be unique to the MB neuropils in the insect brain, and has interesting parallels to the cerebellum in vertebrates [61].

Although we have used whole images in our simulation, we believe the approach is neutral to the issue of whether insects actually use the whole panorama, the skyline, the ground pattern, optic flow, or salient landmarks. Indeed there is no reason why the patterns stored should represent only one modality at a time. The multimodal inputs to the MB in ants [62] suggest that the KC activation pattern might combine olfactory, visual and other sensory (and possibly

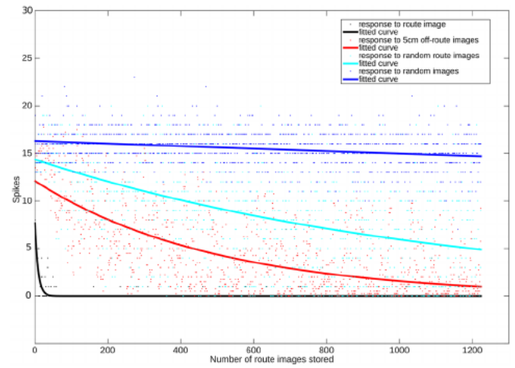


图 6 | 随路线图像持续存入（横轴），检验蘑菇体网络区分熟悉与新图像的容量。平均而言（拟合曲线），EN 对已学图像的输出为零。同向的邻近位置图像比随机位置图像更熟悉（EN 响应更低）。即使存储 1200 幅图像，随机图像仍清楚可辨。

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讨论

我们表明，昆虫蘑菇体的神经结构可以实现蚂蚁循路所需的“熟悉度”算法[11]。在模拟中，蘑菇体从蚂蚁自然栖息地的真实视角重建的、路线上的 80 多幅图像中学习，并可靠地区分这些图像与朝路线外观察时得到的高度相似图像。这是首个与已知神经回路如此紧密对应的昆虫视觉导航模型，而非仅诉诸抽象的计算能力。因此我们发现，比 Infomax 更简单的联想学习网络（计算上等价于 Willshaw 网络[50]）就足以完成这一任务。

值得注意的是，这个最初用于模拟果蝇简单嗅觉联想学习的网络，只需扩大规模、几乎无需其他修改，就能在逼真条件下支持看似复杂得多的导航。已知蚂蚁和蜜蜂等导航昆虫的蘑菇体比果蝇显著增大[57]。昆虫的视觉记忆也曾被定位于脑中的中央复合体[27,58]；导航（包括返巢路线的视觉记忆）完全由中央复合体承担、蘑菇体并不参与，仍是可能的。我们也认为中央复合体很可能是许多昆虫导航能力的关键区域[27,59]，尤其涉及偏振光处理[60]，也可能涉及路径积分。不过，蚂蚁和蜜蜂等膜翅目昆虫要沿长距离路线行进，对记忆的需求极高，可能借用了蘑菇体独特的回路性质：把模式发散投射到由大量神经元承载的稀疏编码中，以区分并存储大量任意且彼此相似的模式。昆虫脑中似乎只有蘑菇体神经毡具备这种结构；它还与脊椎动物的小脑存在值得关注的相似之处[61]。

虽然我们在模拟中使用完整图像，但认为这一方法不偏向于昆虫究竟使用整个全景、天际线、地面纹理、光流还是显著地标这一问题的任何特定答案。所存模式也没有理由一次只能表示一种模态。蚂蚁蘑菇体的多模态输入[62]提示，KC 激活模式可能把嗅觉、视觉和其他感觉（还可能包括

proprioceptive or motivational) inputs into a gestalt experience of the current location, and alteration of any of these cues might reduce the familiarity. Manipulation of olfactory [63], wind [64] and tactile [65] cues have been shown to affect navigational memory in ants.

The network we have described simply stores patterns, rather than trying to learn an underlying generative model to classify 'on-route' vs 'off-route' memories. Nevertheless, in common with many learning problems, the ideal visual memory for route following has an over-fitting/generalization tradeoff. If the memory for individual views is too specific, then any small displacement from the route will result in no direction looking familiar and the animal being lost; and if the similar views resulting from small displacements could be stored as 'one class', the environmental range of a limited capacity memory would be increased. However, if parameters (or the learning rule) are adjusted to allow greater generalization to similar views, there will be increased aliasing and hence risk of mistaking a new view for a familiar one, which would result in moving in the wrong direction, with potentially fatal consequences. One way to deal with this problem might be to introduce more sophisticated visual processing. In particular, a more effective vPN-KC mapping than the random connectivity assumed here might be obtained by allowing self-organisation (unsupervised learning) in response to visual experience in the natural environment [66] so that only 'useful' correlations (for example features that vary strongly with rotation but not displacement [67]) are preserved. Evidence for adaptive synapses at this level of the circuit exists for bees [41] and flies [42]. More complex pre-processing might also be needed to deal with some aspects of the real performance of ants, who continue to follow a route under changing light conditions and movement of vegetation by wind, with some robustness to changing landmark information, and probably also pitch and roll of the head [68,69].

Similarly, we recognise there is likely to be more complexity to the ant's use of PI and visual memory than the algorithm presented here suggests. For example, it would be necessary to postulate a separate memory to explain the outwards routes of ants, with acquisition guided by successful progress along an outbound vector that is expected to lead to a food source. Insects also appear to acquire visual memory during learning walks and flights, by turning back to view the nest [8,70,71]. In this case we would need to assume that the reinforcement signal is under more complex control of a motor programme that maintains or generates fixations in the nest direction. Another possibility is that memory storage is triggered by significant change in the view. Interestingly, this is already inherently determined by the use of a three-factor learning rule in our MB model. If a view is already sufficiently 'familiar', the EN will not fire in response to the KC activation, so the synapses will not be tagged and learning will not occur.

The model we present provides a neurophysiological underpinning for the claim that ants can perform route navigation without requiring a map and for assessing the range over which views can provide guidance. We should be cautious to extend results from ants to other insects, particularly bees, whose sensorimotor habitat is significantly different and might thus have developed different strategies for navigation. However, although it might seem difficult to account for novel routes (or short cuts) ever being taken using the familiarity algorithm, great care is needed to distinguish a truly novel route from the behaviour that may emerge from an animal continuously orienting to make the best visual memory match possible given its current location and stored memories. This can include visual homing within a catchment area larger than that actually explored by the animal [12,72,73]. We also note that though PI is not used in route guidance in our simulations, it will normally still be active in the animal, with potential to influence the direction taken. In particular, recent results have shown that conflicting visual and PI cues often result in a compromise direction being taken by ants [73–77]. Under some circumstances, the result could appear to be a novel shortcut.

本体感觉或动机状态）的输入，整合为对当前位置的整体体验；其中任何线索改变，都可能降低熟悉度。已有研究表明，操纵蚂蚁的嗅觉[63]、风[64]和触觉[65]线索会影响其导航记忆。

我们描述的网络只是存储模式，并不尝试学习一个底层生成模型来区分“路线内”与“路线外”的记忆。然而，与许多学习问题一样，适于循路的视觉记忆必须在过拟合与泛化之间权衡。如果对各个视景的记忆过于精确，稍偏离路线后任何方向都不会显得熟悉，动物便会迷失；若能将小位移产生的相似视景归为“一类”，有限容量的记忆就可覆盖更大的环境范围。但若调整参数或学习规则，使记忆对相似视景更易泛化，不同地点视景混淆的风险也会上升，动物可能把新视景误认成熟悉视景，走向错误方向，甚至导致致命后果。一个应对办法是引入更复杂的视觉处理。例如，让 vPN–KC 映射在自然视觉经验下自组织（无监督学习）[66]，可能比本文假设的随机连接更有效，从而只保留“有用”的相关性，如随旋转明显变化、却较少受平移影响的特征[67]。已有蜜蜂[41]和果蝇[42]的证据表明，这一回路层次存在适应性突触。更复杂的预处理还可能是解释蚂蚁真实表现所必需的：蚂蚁在光照变化、植被被风吹动时仍会循路，对地标信息变化有一定稳健性，而且头部的俯仰与翻滚可能也会变化[68,69]。

同样，我们承认，蚂蚁对路径积分与视觉记忆的使用可能比这里呈现的算法更复杂。例如，要解释蚂蚁外出觅食时的路线，就必须假设存在一套独立记忆，其习得由沿预期通向食物源的外出向量取得进展来引导。昆虫还会在学习步行或学习飞行期间回身朝巢穴观察，从而获得视觉记忆[8,70,71]。对此，我们需要假设强化信号受到更复杂的运动程序控制，由该程序维持或产生朝向巢穴的注视。另一种可能性是，视景发生显著变化时才触发记忆存储。有趣的是，我们蘑菇体模型采用的三因素学习规则本身就具有这一性质：若视景已经足够“熟悉”，KC 激活不会引起 EN 放电，突触便不会被标记，学习也不会发生。

本模型为蚂蚁无需地图也能循路导航这一主张提供了神经生理学基础，也有助于评估视景能够提供方向指引的范围。不过，将蚂蚁的结果推广到其他昆虫，尤其是感觉运动生境显著不同、可能发展出不同导航策略的蜜蜂时，应当谨慎。熟悉度算法似乎难以解释动物何以会采用新路线或捷径；但判断一条路线是否真正“新颖”也需要十分谨慎，因为动物在当前位置和既有记忆的约束下，持续调整朝向以获得最佳视觉记忆匹配，也可能产生看似新路线的行为。这还包括在大于其实际探索区域的吸引域内进行视觉返巢[12,72,73]。此外，虽然我们的模拟没有使用路径积分引导路线，真实动物的路径积分系统通常仍在运行，并可能影响所选方向。近期结果尤其表明，当视觉与路径积分线索冲突时，蚂蚁常选择折中的方向[73–77]；在某些条件下，这可能看起来像一条新的捷径。

Methods

Ant data

The data on ant routes and environment comes from our previous study of route following in *Cataglyphis velox* [43]. Briefly, individual foragers were tracked over repeated journeys, by marking their location on squared paper at approximately 2 second intervals relative to a grid marked on the ground with bottle caps at 1 metre spacing. Subsequently a continuous path was reconstructed using polynomial interpolation. Mapping of tussock locations and panoramic photos from this environment were used to create the virtual environment based on the same software used in [10–11]; details and the simulated world itself are available from www.insectvision.org/walking-insects/antnavigationchallenge.

Image processing

The simulated ant's visual input, for a specific location and heading direction, is reconstructed from a point 1cm above the ground plane, with a field of view that extends horizontally for 296 degrees and vertically for 76 degrees, with 4 degree/pixel resolution, producing a 19x74 pixel image. The greyscale image values are inverted and the local contrast is enhanced using contrast-limited adaptive histogram equalisation, by applying the Matlab function *adapthisteq* [http://uk.mathworks.com/help/images/ref/adapthisteq.html] with default values. It is then further down-sampled to 10x36 pixels using the Matlab function *imresize* [http://uk.mathworks.com/help/images/ref/imresize.html] in which each output pixel is a weighted average of the pixels in the nearest 4x4 neighbourhood. For input into all models, the image is normalised by dividing each pixel value by the square root of the sum of squares of all pixel values. Matlab function *reshape* [http://uk.mathworks.com/help/matlab/ref/reshape.html] is applied to convert the normalised 2D (10x36) image into an 1D (360x1) vector used for subsequent processing.

Spiking neural network

The Mushroom Body network is essentially the same as that described in [30] except that it uses more neurons and has anti-Hebbian learning. It is composed of three layers (see Fig 2 and Tables 1–3) of 'Izhikevich' spiking neurons [78], i.e., for each neuron the change in membrane potential v(mV) is modelled as follows:

$$C\dot{v} = k(v - v_r)(v - v_t) - u + I + [\xi \sim N(0, \sigma)] \quad (2)$$

$$\dot{u} = a(b(v - v_r) - u) \quad (3)$$

The variables v (membrane potential) and u (recovery current) are reset if the membrane potential exceeds a threshold v_t:

$$\begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases} \quad (4)$$

Table 1. Connectivity and synaptic weights.

From X to Y	Prob. of connection	Initial Weight per connection
vPN to KC	Each KC receives input from 10 randomly selected PNs	0.25 (constant)
KC to EN	1 (all KCs connect to the EN)	2.0 (subject to learning)

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方法

蚂蚁数据

蚂蚁路线和环境数据来自我们此前对 Cataglyphis velox 循路行为的研究[43]。简要而言，我们在蚂蚁反复行进的过程中追踪单只觅食蚂蚁，以地面上每隔 1 m 放置的瓶盖构成的网格为参照，大约每 2 秒在方格纸上标记一次蚂蚁位置。随后使用多项式插值重建连续路径。我们根据草丛位置测绘和环境全景照片，采用文献[10,11]使用的软件构建虚拟环境。具体细节和模拟世界见原文给出的数据库网址：www.insectvision.org/walking-insects/antnavigationchallenge。

图像处理

在指定位置和朝向，模拟蚂蚁的视觉输入从离地 1 cm 处重建，水平视野 296°、垂直视野 76°，分辨率每像素 4°，得到 19×74 像素的图像。灰度值反转后，采用对比度受限的自适应直方图均衡化增强局部对比度：使用 Matlab 的 adapthisteq 函数及默认参数。再用 imresize 降采样为 10×36 像素，其中各输出像素为最近 4×4 邻域内像素的加权平均。输入所有模型前，每个像素值除以全部像素值平方和的平方根以归一化。最后用 Matlab 的 reshape，将归一化的二维（10×36）图像转为后续处理使用的一维（360×1）向量。

脉冲神经毡络

除增加神经元数量并采用反赫布学习外，蘑菇体网络与文献[30]描述的网络基本相同。它由三层 Izhikevich 脉冲神经元组成（见图 2、表 1–3）[78]。每个神经元的膜电位 v（mV）的变化建模如下：

$$C\dot{v} = k(v - v_r)(v - v_t) - u + I + [\xi \sim N(0, \sigma)] \quad (2)$$

$$\dot{u} = a(b(v - v_r) - u) \quad (3)$$

若膜电位超过阈值 v_t，变量 v（膜电位）和 u（恢复电流）按下式重置：

$$\begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases} \quad (4)$$

表 1 | 连接关系与突触权重

从 X 到 Y	连接概率	每条连接的初始权重
vPN → KC	每个 KC 接收随机选取的 10 个 PN 的输入	0.25（固定）
KC → EN	1（全部 KC 都连接到 EN）	2.0（随学习改变）

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Table 2. Parameters for each neuron type.

Parameter	PN	KC	EN
Neuron Number	360	20000	1
Scaling factor	5250	N/A	N/A
C	100	4	100
a	0.3	0.01	0.3
b	-0.2	-0.3	-0.2
c	-65	-65	-65
d	8	8	8
k	2	0.035	2
v_r	-60	-85	-60
v_t	-40	-25	-40
ξ	$N(0, 0.05)$	$N(0, 0.05)$	$N(0, 0.05)$

Neuron Number is the number of neurons in each type. Scaling factor is specifically for visual Projection Neurons: the input signal without scaling is the result from normalisation of grayscale images, thus in the range of [0, 1]. The scaling factor amplifies the signal so that any image will activate about half of the vPNs. The parameters from C to v_t in the first column are Izhikevich spiking neuronal model parameters. ξ defines Gaussian random noise current injected into each neuron.

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The parameter C is the membrane capacitance, v_r is the resting membrane potential, I is the input current, $\xi \sim N(0, \sigma)$ is noise with a Gaussian distribution, and a, b, c, d and k are model parameters (see Table 2) which determine the characteristic response of the neuron.

Synaptic input to the first layer, which consists of 360 visual projection neurons (vPNs), is given as an input current proportional to the corresponding pixel in the normalised image 1D vector (360x1) described above:

$$PN\ \ input = \ \ scaling\ \ factor \times Vector$$

The normalization and choice of scaling factor ensures that approximately half the vPNs are activated by each image. The subsequent layers consist of 20,000 Kenyon cells (KC), each receiving input from 10 randomly selected vPNs, and a single extrinsic neuron (EN) receiving input from all KC, and their synaptic input is modelled by:

$$I = gS(v_{rev} - v) \tag{5}$$

Table 3. Parameters for each synapse type.

Parameter	PN to KC	KC to EN
τ_{syn}	3.0	8.0
ϕ	0.93	8.0
g	0.25	[0, 2.0]
τ_c	N/A	40ms
τ_d	N/A	20ms

The parameter τ_{syn} is the time constant for synapses, ϕ represents the quantity of neurotransmitter released per synapse, and g is the weight per synapse. Note that the weights of connections from vPN to KC are fixed whereas the weights from KC to EN are bounded to the range of 0 to 2.0. The parameter τ_c is the time constant for Izhikevich type synaptic eligibility trace in KC-EN synapses, and τ_d is the Biogenic Amine concentration time constant.

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表 2 | 各类神经元的参数

参数	PN	KC	EN
神经元数量	360	20000	1
缩放因子	5250	N/A	N/A
C	100	4	100
a	0.3	0.01	0.3
b	-0.2	-0.3	-0.2
c	-65	-65	-65
d	8	8	8
k	2	0.035	2
v_r	-60	-85	-60
v_t	-40	-25	-40
ξ	N(0, 0.05)	N(0, 0.05)	N(0, 0.05)

神经元数量为各类型细胞的个数。缩放因子专用于视觉投射神经元：未缩放的输入为灰度图归一化结果，取值在 [0,1]；放大后任一图像约激活一半 vPN。首列 C 至 v_t 是 Izhikevich 脉冲神经元模型参数；ξ 表示注入各神经元的高斯随机噪声电流。N/A 表示不适用。

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参数 C 为膜电容，v_r 是静息膜电位，I 是输入电流，ξ~N(0,σ) 表示服从高斯分布的噪声；a、b、c、d、k 是决定神经元特征性响应的模型参数（见表 2）。

第一层由 360 个视觉投射神经元（vPN）构成，其突触输入是与上述归一化图像的一维向量（360×1）中对应像素成正比的输入电流：

PN 输入 = 缩放因子 × 向量

归一化与缩放因子的选择，保证每幅图像大约激活一半 vPN。后续层由 20,000 个 KC（每个接收随机选取的 10 个 vPN 输入）及单个 EN（接收全部 KC 输入）组成；其突触输入建模如下：

$$I = gS(v_{rev} - v) \tag{5}$$

表 3 | 各类突触的参数

参数	PN → KC	KC → EN
τ_{syn}	3.0	8.0
φ	0.93	8.0
g	0.25	[0, 2.0]
τ_c	N/A	40 ms
τ_d	N/A	20 ms

τ_{syn} 为突触时间常数，φ 表示每条突触释放的神经递质量，g 为突触权重。vPN 到 KC 的权重固定，KC 到 EN 的权重限于 0–2.0。τ_c 为 KC–EN 突触中 Izhikevich 型突触资格迹的时间常数；τ_d 为生物胺浓度时间常数。N/A 表示不适用。

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where g (nS) is the maximal synaptic conductance (the synaptic 'weight'), $v_{rev} = 0$ is the reversal potential of the synapses, and S is the amount of active neurotransmitter which is updated as follows:

$$\dot{S} = \frac{-S}{\tau_{syn}} + \phi\delta(t - t_{pre}) \tag{6}$$

where the parameter ϕ is a quantile of the amount of neurotransmitters released when a pre-synaptic spike occurred, τ_{syn} is the synaptic time constant, t_{pre} is the time at which the presynaptic spike occurred and δ is the Dirac delta function. See Table 3 for synapse parameter values. The weights from vPN to KC are fixed. The weights g from KC to EN are altered by learning using a modification of the three-factor rule from [79]:

$$\dot{g} = cd \tag{7}$$

where g is the synaptic conductance, c is a synaptic 'tag' which maintains an eligibility trace over short periods of time, signalling which KC was involved in activating EN (see below); and d is the extracellular concentration of a biogenic amine, modelled by:

$$\dot{d} = \frac{-d}{\tau_d} + BA(t) \tag{8}$$

where $BA(t)$ is the amount of biogenic amine released at time t, which depends on the presence of the reinforcement signal [30] (see training procedure below) and τ_d is a time constant for the decay of concentration d.

The synaptic tag c is modified using spike timing dependent plasticity (STDP):

$$\dot{c} = \frac{-c}{\tau_c} + STDP(t_{pre} - t_{post})\delta[(t - t_{pre}) * (t - t_{post})] \tag{9}$$

where $\delta(t)$ is the Dirac delta function, t_{pre} is the time of a pre-synaptic spike, t_{post} is the time of a post-synaptic spike, and τ_c is a time constant for decay of the tag c. Thus, either pre- or post-synaptic neuronal firing will change variable c by the amount STDP, defined as:

$$STDP(t_{pre} - t_{post}) = \begin{cases} A_+ e^{\frac{t_{pre} - t_{post}}{\tau_+}}, & if\, t_{pre} - t_{post} < 0 \\ 0, & if\ \ t_{pre} - t_{post} = 0 \\ A_- e^{\frac{t_{pre} - t_{post}}{\tau_-}}, & if\ \ t_{pre} - t_{post} > 0 \end{cases} \tag{10}$$

A_+ and A_- are the amplitudes, and τ_+/τ_- are the time constants. However we use an anti-Hebbian form of this rule, such that the tag is always negative:

$$\begin{aligned} A_+ &= A_- = -1.0 \\ \tau_+ &= \tau_- = 15 \end{aligned} \tag{11}$$

Network geometry

Training procedure

We assume that an ant learns a route while running home (following a home vector) and avoiding obstacles. This gives rise to a unique set of visual experiences which it memorises, so that subsequent traversals of the same route can be made without a home vector, and starting

其中 g（nS）为最大突触电导，即突触“权重”；v_{rev}=0 为突触的反转电位，S 为活跃神经递质的量，按下式更新：

$$\dot{S} = \frac{-S}{\tau_{syn}} + \phi\delta(t - t_{pre}) \tag{6}$$

参数 φ 表示突触前脉冲发生时释放的神经递质量，τ_{syn} 为突触时间常数，t_{pre} 为突触前脉冲发生的时刻，δ 为狄拉克 δ 函数。突触参数见表 3。vPN 到 KC 的权重固定；KC 到 EN 的权重 g 通过对文献[79]三因素规则的修改而在学习中改变：

$$\dot{g} = cd \tag{7}$$

其中 g 是突触电导；c 是突触“标记”，短时间保留资格迹，指示哪个 KC 参与激活 EN（见下文）；d 是细胞外生物胺浓度，建模如下：

$$\dot{d} = \frac{-d}{\tau_d} + BA(t) \tag{8}$$

BA(t) 是时刻 t 释放的生物胺量，取决于是否存在强化信号[30]（见下文“训练程序”）；τ_d 为浓度 d 衰减的时间常数。

突触标记 c 通过脉冲时序依赖可塑性（STDP）修改：

$$\dot{c} = \frac{-c}{\tau_c} + STDP(t_{pre} - t_{post})\delta[(t - t_{pre}) * (t - t_{post})] \tag{9}$$

其中 δ(t) 为狄拉克 δ 函数，t_{pre} 和 t_{post} 分别为突触前、后脉冲的时刻，τ_c 为标记 c 衰减的时间常数。因此，突触前或后神经元放电时，变量 c 都按以下定义的 STDP 值改变：

$$STDP(t_{pre} - t_{post}) = \begin{cases} A_+ e^{\frac{t_{pre} - t_{post}}{\tau_+}}, & if\, t_{pre} - t_{post} < 0 \\ 0, & if\ \ t_{pre} - t_{post} = 0 \\ A_- e^{\frac{t_{pre} - t_{post}}{\tau_-}}, & if\ \ t_{pre} - t_{post} > 0 \end{cases} \tag{10}$$

A_+、A_- 是幅度，τ_+、τ_- 是时间常数。不过，我们使用该规则的反赫布形式，使标记始终为负：（译注·存疑：式（10）正时差分支的指数疑缺负号，与图 3B 的衰减趋势不一致；尚待核实。）

$$\begin{aligned} A_+ &= A_- = -1.0 \\ \tau_+ &= \tau_- = 15 \end{aligned} \tag{11}$$

网络结构

训练程序

我们假设，蚂蚁避开障碍、沿返巢向量回巢时学会路线。这产生一组被记住的独特视觉体验，使其此后无需返巢向量，也能沿同一路线行进，并可从路线上的

from any point along it. In practice, we take each recorded route of a real ant (of average length 8m) and use it to train the spiking neural network as follows. From nest to home, every 10cm along the route, we use the simulated environment to generate an image facing the next 10cm waypoint. After pre-processing as described above, the normalised image pixel values are given as input to the first layer (vPNs) of the network every 1ms for 40ms. Then a reinforcement signal is presented for one time-step (BA(t) = 0.5, for t = 40ms from image onset) and a further 10ms of simulation time (with no image or reinforcement signal presented) allowed to elapse, during which any synaptic weight changes will occur. Given the parameters below, this presentation results in rapid, 'one-shot' learning of the presented image, with weights between any active KC and EN decreasing rapidly to zero (see Fig 3). To save computation time, we do not explicitly model network activity during the lapse of time until the next image to be learnt would be encountered (during which the ant moves 10cm further along the route). Instead we simply reset the membrane potentials, synaptic tags and amine levels in the network to their initial state, and then present the new image for 40ms, followed by reinforcement, etc., until all the images for that route (around 80 for an 8m route) have been presented. Subsequently the spiking rate output of EN will indicate the novelty of any image presented to the network.

Perfect memory

As a benchmark for the difficulty of the navigation task, we determine the performance of a simulated ant which has a perfect memory, i.e., it simply stores the complete set of training images, and uses direct pixel-by-pixel image differencing to compare images [9]. When recapitulating a route, novelty of the current view I with respect to all stored images is calculated as:

$$Novelty(I) = \min_i \left(\sum_{x,y} (I(x,y) - V_i(x,y))^2 \right) \quad (12)$$

Where V_i is each of the stored images, and x,y define the individual pixels.

Infomax

For comparison, we also implemented the Infomax algorithm, a continuous 2-layer network, closely following the method described in [11], except with a much higher learning rate. The input layer has $N = 360$ units and the normalized intensities of the 36x10 image pixels are mapped row by row onto the input layer to provide the activation level of each unit, x_i. The output layer also has 360 units and is fully connected to the input layer. The activation of each unit of the output layer, h_i, is given by:

$$h_i = \sum_{j=1}^{N} w_{ij} x_j \quad (13)$$

Where w_{ij} is the weight of the connection from the *jth* input unit to the *ith* output unit. The output y_i is then given by:

$$y_i = \tanh(h_i) \quad (14)$$

Each image is learnt in turn by adjusting all the weights (initialized with random values) by:

$$\Delta w_{ij} = \frac{\eta}{N} (w_{ij} - (y_i + h_i) \sum_{k=1}^{N} h_k w_{kj}) \quad (15)$$

Where $\eta = 1.1$ is the learning rate.

任一点出发。实际训练采用真实蚂蚁的各条记录路线（平均长 8 m）。原文写“从巢穴到家”，沿路线每隔 10 cm，用模拟环境生成朝向下一个 10 cm 路点的图像。（译注·已核实：原文“from nest to home”与返巢训练设定不一致；按结果部分及图 1，此处应指从食物源沿路线返回巢穴。保留原文表述供核对。）按上述方式预处理后，在 40 ms 内每隔 1 ms，将归一化像素值输入网络第一层 vPN。随后在一个时间步给予强化信号（从图像开始呈现计，t=40 ms 时 BA(t)=0.5），再运行 10 ms；此时不呈现图像或强化信号，突触权重可在此期间改变。按下述参数，这一过程使所呈图像被迅速“一次学会”，活跃 KC 到 EN 的权重快速降至零（图 3）。为节省计算，我们不显式模拟遇到下一幅待学图像前的网络活动，即蚂蚁沿路线再走 10 cm 的过程，而将膜电位、突触标记和胺水平重置为初始状态，再呈现新图像 40 ms、随后强化，依次类推，直到该路线全部图像（8 m 约 80 幅）呈现完毕。此后，EN 输出放电频率即可指示网络输入图像的新颖程度。

完美记忆

为衡量导航任务的难度，我们确定具有完美记忆的模拟蚂蚁的表现：它存储全部训练图像，通过逐像素直接作差比较图像[9]。重走路线时，当前视景 I 相对于全部已存图像的新颖度计算如下：

$$Novelty(I) = \min_i \left(\sum_{x,y} (I(x,y) - V_i(x,y))^2 \right) \quad (12)$$

V_i 为每一幅已存图像，x、y 表示各像素坐标。

Infomax

作为比较，我们也实现了 Infomax 算法。该连续双层网络基本遵循文献[11]，但学习率高得多。输入层含 N=360 个单元，36×10 图像像素的归一化亮度逐行映射为各输入单元的激活值 x_i。输出层也含 360 个单元，与输入层全连接。每个输出单元的激活值 h_i 为：

$$h_i = \sum_{j=1}^{N} w_{ij} x_j \quad (13)$$

w_{ij} 是第 j 个输入单元到第 i 个输出单元的连接权重；输出 y_i 为：

$$y_i = \tanh(h_i) \quad (14)$$

依次学习各图像，按下式调整所有权重（初始为随机值）：

$$\Delta w_{ij} = \frac{\eta}{N} (w_{ij} - (y_i + h_i) \sum_{k=1}^{N} h_k w_{kj}) \quad (15)$$

其中学习率 η=1.1。

Subsequently the novelty of an image is given by the summed activation of the output layer:

$$d(\vec{x}) = \sum_{i=1}^{M} | h_i | \tag{16}$$

Capacity

For this analysis we simplify the MB model by assuming the KC have a binary state (activated if input is above threshold, otherwise not active) and learning will alter the state of the respective KC-EN synapse by setting it to zero when KC activity coincides with reward. We can analyse the storage capacity of a network with N input nodes and average activity p by deriving an expression for the probability of error P_{error}, defined as the probability that a random unlearned pattern activates only KC nodes that already have their weight set to 0, producing the same EN output as a learned pattern (i.e. 0). We assume the number of neurons k coding a pattern is drawn from the binomial distribution with probability p. After learning m random patterns:

$$P(w_i = 0) = 1 - (1-p)^m \tag{17}$$

The probability that a new pattern activates only neurons with $w_i = 0$ is given by:

$$P_{error} = \sum_K \binom{N}{k} (p(1-(1-p)^m))^k (1-p)^{N-k} \tag{18}$$

That is, the sum of possible arrangements of k active units that all have weights set to 0. This can be written as:

$$\begin{aligned} P_{error} &= (1-p)^N \Sigma_k \binom{N}{k} \left(\frac{p(1-(1-p)^m)}{1-p}\right)^k \\ &= (1-p)^N \left(1 + \frac{p(1-(1-p)^m)}{(1-p)}\right)^N \\ &= (1-p(1-p)^m)^N \end{aligned} \tag{19}$$

For a given acceptable error rate P_{error}, we can solve for the number of patterns m that can be stored before that error rate will be exceeded:

$$\mathrm{m} = \frac{\ln\left(\frac{1-P_{error}{}^{1/N}}{p}\right)}{\ln(1-p)} \tag{20}$$

Fig 5 shows how the capacity m of the network changes with the number of neurons N and average activity p, for $P_{error} = 0.01$.

To confirm this analysis is consistent with the behaviour of the full spiking network, we carried out an equivalent capacity testing process using the simulation. For a network of a given size, we generated random activation of the KC neurons at different levels of average activity. Patterns were learned as before, i.e., by applying a reward signal after 40ms, resulting in altered KC-EN weights. After each successive pattern was learned, we tested the network with 100 random patterns, to see if any produced a spiking response as low as that of the learned patterns, i.e., would potentially be confused with a learnt pattern (see S2 Fig for the result with N = 20000, p = 0.01). We noted how many patterns could be learnt before $P_{error} = 0.01$, i.e. where ≥ 1 out of 100 new patterns would start to be confused with a learnt pattern.

Note however this analysis and simulation assumes both learned and tested KC patterns are independent and random, which is not strictly true in the navigation task. We also tested the

此后，图像的新颖度由输出层的激活值求和给出：

$$d(\vec{x}) = \sum_{i=1}^{M} | h_i | \tag{16}$$

容量

为进行分析，我们简化蘑菇体模型：KC 为二值状态，输入超过阈值时激活，否则不激活；学习在 KC 活动与奖赏同时出现时，将相应 KC–EN 突触权重设为零。对含 N 个输入节点、平均活动比例 p 的网络，推导错误概率 P_{error} 以分析容量。“错误”指未学习的随机模式只激活权重已为零的 KC，产生与已学习模式相同的 EN 输出（即零）。假设编码某个模式的神经元数 k 服从概率参数为 p 的二项分布。学会 m 个随机模式后：

$$P(w_i = 0) = 1 - (1-p)^m \tag{17}$$

新模式只激活 w_i=0 的神经元的概率为：

$$P_{error} = \sum_K \binom{N}{k} (p(1-(1-p)^m))^k (1-p)^{N-k} \tag{18}$$

即对 k 个活跃单元的所有可能排列求和，这些单元的权重均为零。可写为：

$$\begin{aligned} P_{error} &= (1-p)^N \Sigma_k \binom{N}{k} \left(\frac{p(1-(1-p)^m)}{1-p}\right)^k \\ &= (1-p)^N \left(1 + \frac{p(1-(1-p)^m)}{(1-p)}\right)^N \\ &= (1-p(1-p)^m)^N \end{aligned} \tag{19}$$

给定可接受错误率 P_{error}，可以解出达到该错误率前可存储的模式数 m：

$$\mathrm{m} = \frac{\ln\left(\frac{1-P_{error}{}^{1/N}}{p}\right)}{\ln(1-p)} \tag{20}$$

图 5 显示 P_{error}=0.01 时，网络容量 m 随神经元数 N 和平均活动比例 p 的变化。

为了验证上述分析与完整脉冲网络的行为一致，我们用模拟进行等效的容量测试。对给定规模的网络，我们在不同平均活动水平下，随机激活 KC 神经元。模式仍按之前的方法学习：呈现 40 ms 后施加奖赏信号，使 KC–EN 权重改变。每学会一个新模式后，我们都以 100 个随机模式测试网络，观察其中是否有模式的放电反应低至与已学习模式相同，从而可能与已学习模式混淆（N = 20,000、p = 0.01 的结果见补充图 S2）。我们记录达到 P_{error} = 0.01 之前能够学习多少模式，也就是每 100 个新模式中开始出现至少 1 个与已学模式混淆之前的数量。

但这一分析和模拟假定已学习与测试的 KC 模式彼此独立、随机；导航任务并不严格如此。我们还测试了

spiking network (with N = 20000, p = 0.01) by continuously storing additional patterns generated by real route images, and counting the spikes produced by EN for the learnt image, for an image taken from a 5cm displacement facing the same way, for an image taken at a random location in the ant environment, and for a random image. The results are shown in Fig 6.

External database

The virtual ant world is available from: www.insectvision.org/walking-insects/antnavigationchallenge.

Supporting Information

S1 Movie. This movie shows the result of a route re-capitulation by a simulated ant using the mushroom body model, and the corresponding visual information from the ant's point of view that is used as input to the model. The direction of each 10cm step along the route is chosen by scanning (the scanning movement is not shown) ±60 degrees and choosing the direction producing the fewest spikes from the extrinsic neuron output. The simulated ant never departs far enough from the trained route to get lost, and successfully returns home over 8 meters in the complex environment.
(ZIP)

S1 Fig. A. Distant view of the study site on the outskirts of Seville, Spain with the approximate nest position indicated by the arrow. The nest is surrounded by grass shrub blocking the view of distant objects such as trees. B. Close up view of the field site with the ant nest and experimental feeder marked. C. An example route followed by an ant through the environment shown in blue from an overhead perspective. D. Panoramic images sampled along the route are shown which clearly demonstrate that distant objects were not visible to homing ants.
(EPS)

S2 Fig. Capacity of a MB network with N = 20000 and p = 0.01. From Fig 5, the abstracted model provides the estimate that around 375 random images can be stored (KC weights set to 0) before the probability of an error (a new random image activates only KCs that have already had weights set to 0) exceeds 0.01. Using the full spiking network and the three factor learning rule, we train successively with 500 random KC activation patterns. After each additional pattern is stored, we test the network with 100 random patterns to see how many produce an error (have an EN output of 0 spikes, indicating a familiar pattern). More than 350 patterns could be stored before $> 1/100$ errors occur. The same method is used to generate data points for other values of N and p plotted in Fig 5.
(EPS)

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Author Contributions

Conceived and designed the experiments: BW MM PA FP. Performed the experiments: PA FP MM. Analyzed the data: PA FP MM KL BW. Wrote the paper: BW PA FP MM KL.

N=20,000、p=0.01 的脉冲网络：连续存储真实路线图像产生的模式，并分别统计 EN 对已学图像、相距 5 cm 且朝向相同的图像、蚂蚁环境随机位置的图像及随机图像产生的脉冲。结果见图 6。

外部数据库

虚拟蚂蚁环境见：www.insectvision.org/walking-insects/antnavigationchallenge。

补充材料说明

S1 视频 | 模拟蚂蚁使用蘑菇体模型重走路线的结果，以及从蚂蚁视角输入模型的相应视觉信息。蚂蚁每次沿路线行进 10 cm 前，在 ±60° 内扫描（视频未显示扫描动作），选择使 EN 输出脉冲最少的方向。它从未偏离训练路线远到迷失，在复杂环境中走过 8 m，成功返回巢穴。（ZIP 文件）

S1 图 | A：西班牙塞维利亚郊区研究地点的远景，箭头示巢穴大致位置；巢穴周围草灌丛挡住了树木等远处物体。B：近景，标明蚁巢和实验食物源。C：该环境中蚂蚁所循路线的示例，从俯视角以蓝线表示。D：沿该路线取样的全景图像，清楚显示返巢蚂蚁看不到远处物体。（EPS 文件）

S2 图 | N=20,000、p=0.01 的蘑菇体网络容量。图 5 中的抽象模型估计：可存储约 375 幅随机图像（KC 权重设为零），之后错误概率才超过 0.01；错误指新随机图像只激活权重已为零的 KC。采用完整脉冲网络和三因素规则，依次训练 500 个随机 KC 活动模式。每存入一个额外模式，以 100 个随机模式测试，统计其中有多少产生错误（EN 输出为零脉冲，表示熟悉模式）。在错误数超过 1/100 前，可存储 350 多个模式。图 5 中其他 N、p 取值的数据点也按同一方法生成。（EPS 文件）

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作者贡献

构思与设计：BW、MM、PA、FP；实验：PA、FP、MM；数据分析：PA、FP、MM、KL、BW；论文撰写：BW、PA、FP、MM、KL。

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译者阅读分析 | 研究问题与机制

非原文译文 · 阅读分析 1/4　　左侧完整保留原文书目；以下分析依据前文正文、图表与方法。

核心问题：微小的脑如何记住复杂路线？ 译者梳理

本文研究蚂蚁能否依靠沿途视觉的“熟悉度”选择方向，而无需在每一步先恢复自身位置或建立显式地图。

作者把野外记录路线与重建的视觉场景送入蘑菇体脉冲网络，检验一个原用于嗅觉联想的回路能否承担视觉路线记忆。贡献在于把算法、神经结构和较真实的任务条件联系起来；模型成功支持计算可行性，并不直接证明蚂蚁实际采用了全部设定。

定位：原文第2–3页；第11页讨论。

信息流：图像 → 稀疏编码 → 熟悉度输出 译者梳理

输入：图像预处理为10×36像素，映射到360个视觉投射神经元（vPN）。

扩展：20,000个肯扬细胞（KC）各接收随机10个vPN输入；每幅图像约激活200个KC，形成稀疏活动。

读出：KC汇聚到一个外在神经元（EN）。图像与强化配对时，三因素反赫布规则减弱已标记的KC–EN突触；之后熟悉图像引起较低的EN放电。此处“更熟悉”对应输出更低，而不是更高。

定位：第4–5页输入及模型说明；图2–3。

从记忆到行动：扫描后选择最低输出方向 作者假设与解释

训练沿约8 m记录路线，每10 cm取一幅朝向下一路点的图像；约80幅图像各呈现40 ms，再给予强化。一次配对便可使相关权重接近零，这是模型设定的学习时序。

测试在当前朝向的±60°内扫描，选择EN活动最低的方向，前进10 cm后重复。因此，网络提供熟悉度评估，扫描与行动规则负责把它转成运动。

返巢向量缩短可能提供强化信号，是作者的机制假设；本文未直接记录这种神经强化信号。

定位：第5–7页；第15–16页训练程序。

阅读导引与分析来源 阅读建议 · 尚未执行

接下来三页依次看：测试成绩及其条件 → 随机模式与真实图像的容量区别 → 模型边界和优先核查建议。

分析框架参考 norman-paper-reading（N同学）；分析由本翻译流程生成，未声称运行第三方程序。原文定位均使用本PDF文件页码，不是另加的分析页码。

框架：github.com/N-tongxue/norman-skill

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译者阅读分析 | 循路证据怎样读

非原文译文 · 阅读分析 2/4　　既看成绩，也看评价协议和比较对象。

任务真实性：用记录路线重建视觉体验 译者梳理

野外记录来自Cataglyphis velox，随后重建灌草环境及贴近地面的低分辨率视觉输入。检验对象是模拟网络在这些重建场景上的表现，并非在活体蚂蚁中直接测量该网络的记忆。

这比任意随机图像更接近自然导航的困难：视景相似、地标受遮挡、位置稍变图像就会变化。但重建仍是模型，不能自动涵盖光照、风、头部姿态等全部真实变化。

定位：第3–5页；图1；第12页讨论。

图4的主要结果：误差与计算成本 译者梳理

15条约8 m路线的平均错误数±标准差：随机选向18.7±3.6；完美记忆1.1±0.9；蘑菇体模型2.6±1.5；Infomax 1.5±0.8。

蘑菇体模型明显优于随机基线，但不应称为最佳方法：完美记忆的错误数更低。作者强调的是用较简单、较有生物学依据的学习与读出机制，取得接近比较方法的任务表现，而不是在所有指标上胜出。

定位：第7–8页结果；图4。

关键条件：越界记错，并把智能体移回路线 分析者判断

若一步使模拟蚂蚁偏离路线超过20 cm，就记一次错误，并把它放回路线最近点，再继续测试；终点是进入巢穴20 cm范围。

因此，“平均2.6次错误”是在带移回机制的评价协议下得到的结果。它不能直接解释为完全无辅助、任意离路线后均能自主恢复的成功率；也不能由这一指标直接推出长时间自主导航的可靠性。

定位：第6–7页循路表现评估。

比较设计：每个基线回答不同问题 分析者判断

完美记忆逐一比较当前图像和全部已存图像，衡量熟悉度策略在同一任务中的表现基准。Infomax是另一种网络实现，作者提示其需要更大学习率，并可能过拟合。

分析者判断：精度、连接数量、局部学习规则和读出成本需要分开比较。本文没有因此完成对所有导航算法或真实神经耗能的全面优劣评定。

定位：第5–6、8页；第16–17页比较方法。

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译者阅读分析 | 容量与泛化

非原文译文 · 阅读分析 3/4　　不要把不同定义的“容量”混为一个上限。

容量公式回答什么问题？ 译者梳理

抽象模型把KC和突触简化为二值状态。学习把参与模式的KC–EN权重设低；若一个未学随机模式仅激活这些已减弱连接的KC，输出也低，就会发生“新图像被误认成熟悉”的错误。

容量m描述错误概率达到指定阈值前可存的模式数，依赖KC数N、平均活动比例p及随机模式假设。它不是无条件的“脑能记住多少张照片”，也不是沿所有环境可行走的最大距离。

定位：第9页；第17页容量推导及原始公式。

约375个模式与1200幅路线图像为何并存？ 分析者判断

在N=20,000、p=0.01、允许错误概率0.01的条件下，抽象估计约为375个独立随机模式。另一项测试连续学习1200幅相关路线图像，图6显示平均响应仍能区分已学、邻近和随机位置图像。

两者的输入统计、测试对象和评价方式不同：后一结果并不推翻前者，也不能解读为1200幅图像全都零错误地保持独立可辨。原文明确指出单次EN反应噪声大，仍会偶发混淆。

定位：第9–11、17–18页；图5–6。

稀疏性为何重要，又不能解决什么？ 分析者判断

稀疏活动减少每次学习涉及的连接及模式间干扰，帮助存储许多相似输入。图5把完整脉冲网络的模拟与抽象容量预测比较，支持这种机制在测试设定下的一致性。

但“更稀疏”不自动意味着任意真实场景都更好：还需足够激活、可读出的熟悉度梯度和对小位移的适当泛化。单看容量曲线，不能替代闭环循路评价。

定位：第5、9–10、12页；图5。

泛化的两面：不易迷失，也可能认错 分析者判断

过于精确的视景记忆会让轻微偏离后所有方向都“不熟悉”；过度泛化又会把错误位置或方向误认为熟悉。作者由此讨论视觉预处理、自组织映射及更复杂感觉变化。

阅读重点应是：对邻近正确视景的容忍，能否与对错误方向的区分同时保留，而非仅把熟悉图像的输出压到最低。

定位：第10–12页；图6。

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译者阅读分析 | 边界与下一步

非原文译文 · 阅读分析 4/4　　区分模型可能性、作者假设和待执行建议。

尚未直接证明的生物学环节 分析者判断

论文说明蘑菇体式回路可以实现此类路线学习，但没有直接记录蚂蚁中相同的视觉KC编码、EN熟悉度读出或返巢强化信号。作者也承认视觉路线记忆由中央复合体承担的可能性尚未排除。

因此，仿真可行性、跨昆虫的解剖/生理线索和本物种的因果证据应分开。推广到蜜蜂等感觉运动条件不同的物种，还需额外检验。

定位：第5、11–12页。

原文疑点与未读范围 分析者判断

第9页p_kc被写作连接概率，却用于活跃KC数的期望，存在符号定义疑点；同页350个模式与37.5 m的换算不一致。第15页式（10）的指数符号与图3B趋势可能不符。对应句末已分别附核实或存疑译注，原文主张及公式仍保留。

文件内补充图、视频的文字说明已读；外部补充图S1、S2和视频S1本体未完整取得，不能声称本次已逐项检查这些附件。

定位：第9、15、18页及相应译注。

优先核查建议（尚未执行） 阅读建议 · 尚未执行

先在相同视景和路线条件下测试：KC是否保留足够区分度，EN响应是否随学习降低，并随朝向改变形成可用于选向的熟悉度差。对照应区分新旧图像、正确与错误方向，以及有无强化。

模型层面可优先取消“移回路线”，分别报告到巢成功、偏离恢复和路径误差；再加入光照或植被变化。这样能直接检查现有成绩是否依赖辅助评价条件，而非仅继续优化平均错误数。

依据：第5、7、12页机制与局限。

建议阅读顺序 阅读建议 · 尚未执行

先看图1理解任务，再看图2–3理解稀疏编码与学习，结合图4和第7页评价协议判断循路成绩；最后对照图5–6及第17页推导，区分理论容量和相关图像测试。

可迁移的思想是“稀疏表征 + 快速局部学习 + 熟悉度选向”。它与地图、路径积分或其他导航机制如何共同工作，仍需按具体条件讨论。

分析框架参考 norman-paper-reading；分析由本流程生成。